



Convolutional factor analysis model with application to radar automatic target recognition

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ABSTRACT

Convolutional networks, such as convolutional neural networks, are extensively applied in pattern recognition systems, where the convolution kernel plays an important role in performance improvement. Inspired by this, we integrate the convolution operation into statistical modeling in this paper and develop a novel probabilistic generative model called convolutional factor analysis (CFA), which is more applicable to statistical recognition with limited training data. And then, the CFA model is applied to radar automatic target recognition based on high-resolution range profile (HRRP), where sufficient training data are frequently unavailable due to the sampling rate limitation of real radar systems. As a dictionary learning method, the dimension of each dictionary atom in our CFA model is much lower than that in the traditional factor analysis (FA) model. Meanwhile, the model also makes it possible to capture the basic structures of observed data, thus requiring a fewer dictionary atoms to describe observations. Due to these two properties, the dictionary size of the CFA model is much smaller. Therefore, compared to the traditional FA model, the CFA model has the lower order of model complexity and can be learned better under a small amount of training data. In addition, owing to the conjugate property, the model parameters can be inferred via variational Bayesian (VB) algorithm, and the commutative law of convolution operation is also exploited to simplify the derivations of the posteriors. Experiments on synthetic and measured HRRP data show that basic structures of data can be represented by the dictionaries learned via our CFA model, and the better recognition performance can also be achieved by our method with small training data size.

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1. Introduction

It is known that networks including convolutional restricted Boltzmann machines (RBMs) [1–3], convolutional neural networks (CNNs) [4–7], and deconvolutional networks [8,9], have been widely used in speech recognition and optical image recognition. Compared to fully connected networks, one of the key characteristic of these algorithms is the exploitation of the convolution operator, which is critical in reducing the number of weights and the network complexity. Naturally, we hope to introduce convolution-based methods into radar automatic target recognition (RATR) community, an important branch of pattern recognition area, to obtain excellent recognition performance in the case of limited training samples.

Automatic target recognition (ATR) is a crucial function of the modern radar. Most studies on ATR in high-resolution radar

systems mainly focus on synthetic aperture radar (SAR) ATR [10–13] and high-resolution range profile (HRRP) ATR [14–27,30–33]. Compared to SAR image, radar HRRP can be formed and processed much more easily. Therefore, HRRP-based RATR has elicited considerable attention.

For high-resolution radar system, since the target size is much larger than the wavelength of radar signal, the electromagnetism characteristic of the target can be described by the scattering center model [28,29]. According to this model, a target does not appear as a “point target” anymore but consists of many scatterers distributed in several range cells along the radar line-of-sight (LOS), as illustrated in Fig. 1. Assuming that the radar transmitted signal is $s(t)e^{j\omega_c t}$ ($s(\cdot)$ is the signal envelope and ω_c is the carrier angular frequency), the n th returned echo of the l th scatterer from the p th range cell is

$$x_{npl}(t) = \sigma_{pl} s\left(t - \frac{2R_{npl}}{c}\right) \exp\left\{-j\left[\frac{4\pi}{\lambda} R_{npl} - \psi_{pl0}\right]\right\}, \quad (1)$$

where λ denotes the radar wavelength; R_{npl} denotes the radial distance between the radar and the l th scatterer in the p th range cell

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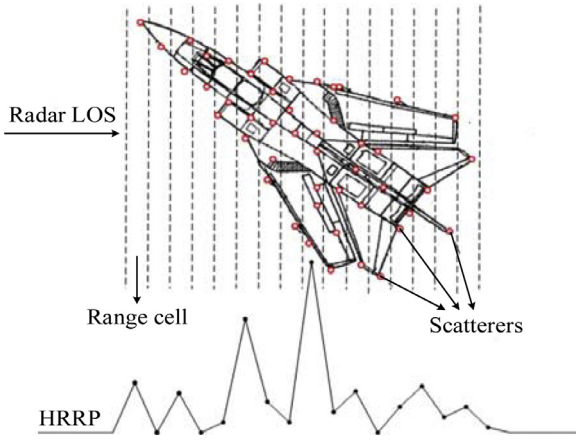


Fig. 1. Illustration of an HRRP sample from a plane target, where the circles on the plane represent the scatterers. This figure is cited from [24].

of the n th returned echo; σ_{pl} and ψ_{pl0} represent the strength and the initial phase of the l th scatterer in the p th range cell, respectively. From (1), we know that the returned echo of each scatterer in a range cell is a complex value related to time, which is called the complex time return of the scatterer. Then, the n th returned echo from the p th range cell $x_{np}(t)$ is defined as the coherent summation of the complex time returns from target scatterers in the p th range cell, that is, $x_{np}(t) = \sum_{l=1}^{V_p} x_{npl}(t)$, where V_p denotes the number of target scatterers in the p th range cell.

Given that the signal envelope $s(\cdot)$ in a range cell is approximately unvaried for high-resolution radar, $x_{np}(t)$ can be approximated as

$$x_{np}(t) \approx x_{np} = \sum_{l=1}^{V_p} \sigma_{pl} \exp \left\{ -j \left[\frac{4\pi}{\lambda} R_{npl} - \psi_{pl0} \right] \right\}. \quad (2)$$

And for the n th HRRP $\mathbf{x}_n$, it can be expressed as $\mathbf{x}_n = [|x_{n1}|, |x_{n2}|, \dots, |x_{np}|]^T$ with $|\cdot|$ and $[\cdot]^T$ denoting modular arithmetic and transpose operation, respectively, that is, an HRRP is the amplitude of the coherent summation of the complex time returns from target scatterers in each range cell.

In statistics and machine learning, statistical recognition method based on Bayes theorem is an efficient approach for HRRP-based target recognition, as demonstrated in [17,19–21,23,24,26,31,33,34]. According to the scattering center model, an HRRP sample is highly sensitive to target-aspect variation, and thus the target-aspect sensitivity should be solved before applying HRRP to statistical recognition. In [21], we proposed the use of a statistical model to describe each aspect-frame, where the HRRPs are collected from a target-aspect sector roughly without the scatterers' motion through range cells (MTRC). By statistical recognition we mean that the class membership of a measurement sample $\mathbf{x}$ is determined based on the posterior probabilities of all frames from all class memberships $\{p(c, m|\mathbf{x})\}_{c=1, m=1}^{C, M_c}$, where $c = 1, 2, \dots, C$ denotes the class membership, C denotes the number of classes, $m = 1, 2, \dots, M_c$ denotes the frame membership, and M_c denotes the number of the aspect-frames in the c th class. According to Bayes theorem, we have $p(c, m|\mathbf{x}) \propto p(\mathbf{x}|c, m)p(c, m)$, where $p(c, m)$ is the prior frame probability and $p(\mathbf{x}|c, m)$ is the frame-conditional likelihood of the m th frame for target c . Since the prior frame probability is commonly assumed to be uniformly distributed, the estimation of the posterior probability is transformed into the estimation of the frame-conditional likelihood.

Some parametric statistical models have been developed to describe the frame-conditional likelihood of each aspect-frame HRRP data. The Adaptive Gaussian Classifier (AGC) model [27], Gaussian model [20], Gamma model [19], and the two-distribution com-

pounded model comprising Gamma distribution and Gaussian mixture distribution [26] were proposed with the hypothesis that the elements in an HRRP sample are statistically independent. However, the independent assumptions weaken the data description capabilities of these models and limit their recognition accuracies. Motivated by this, scholars developed several more accurate statistical models, such as subspace approximation model [21], probability principal components analysis (PPCA) model [21,36], factor analysis (FA) model [21], and the local factor analysis (LFA) model [24], that consider the statistical correlations among different range cells in an HRRP sample. Nevertheless, these models have the higher order of model complexity and require substantial training data to estimate the model parameters accurately. Due to the sampling rate limitation of real radar systems, it is hard to collect enough training data for each target-aspect of each target. Instead of building an individual model for each data subset associated with each target-aspect, the thought of sharing the information among all related data, called multi-task learning (MTL) [35], was applied to HRRP-based RATR for improving the recognition performance with limited training data, e.g., multi-task learning factor analysis (MTL-FA) model [17] and multi-task learning-based hidden Markov model with truncated stick-breaking prior (MTL-TSB-HMM) [33,34]. However, MTL-based statistical models possess several disadvantages. First, these models require the huge computation burden, massive storage space, and excessive training time for the target recognition with a large number of target classes. Second, all the model parameters of all targets need to be relearned when the data of a new target are obtained for training. Although the online learning method [33] can avoid this complicated training phase, its recognition accuracy is lower than that of the relearning method.

As demonstrated at the beginning of this section, convolutional networks have a fewer parameters due to the introduction of convolution kernels. Inspired by this, we introduce the convolution operator into FA model, a Bayesian form of dictionary learning, to depict data via the convolution of dictionary atoms and weight vectors rather than simple linear projection. As a convolution kernel, each atom of the dictionary in our method has lower dimension and tends to capture the basic structure of the data, leading to the fact that fewer dictionary atoms are required compared with the FA model. If we refer to the FA model exploiting the convolution operation as the convolutional factor analysis (CFA) model, analysis above shows that the dictionary size of the CFA model should be much smaller than that of the FA model. Consequently, the CFA model has fewer parameters to be estimated and can be learned better with small training data size, thereby achieving more inspiring recognition performance. A previous work [9] also integrated the convolution operation into the traditional FA model and proposed a hierarchical convolutional factor analysis based on deep learning (DL-HCFA). However, the method in [9] remarkably differs from ours. The DL-HCFA model focuses on extracting deep features in the context of big data, and then a classifier, such as nonlinear support vector machine (SVM), is independently trained using the learned features. By contrast, our CFA model mainly concentrates on the description of observations' statistical characteristics under the limited training data condition, and moreover, the Bayesian classifier is constructed based on the learned model parameters, which avoids the mismatch of the features and the back-end classifier occurred in the DL-HCFA model. In addition, the modeling method in [9] is unsupervised, whereas we indirectly use the supervised information of data by constructing a CFA model for each aspect-frame of each target, thus helping to achieve better recognition performance.

The main contributions of this paper are summarized as follows:

- Inspired by the convolutional networks, we introduce the convolution operation into a probabilistic model and propose a novel generative model, i.e., CFA model, in which dictionary atoms tend to extract the basic structures of data. Furthermore, we combine the proposed convolutional model with statistical recognition and apply it to RATR.
- Compared to the traditional FA model, our CFA model has a fewer parameters to be estimated and is more applicable to small sample recognition problem. This property of the CFA model is practically significant for RATR, where training data is often insufficient due to the sampling rate limitation of real radar systems.
- We directly exploit the commutative law of the convolution operation when estimating the posteriors of model parameters via variational Bayesian (VB) algorithm, which simplifies the derivation of dictionary or weight vectors.

The remainder of the paper is organized as follows. After a brief review of the FA model, we introduce our CFA model in Section 2. Then the model learning and classification method are discussed in Section 3. Detailed experiments are conducted on synthetic and measured data in Section 4. Finally, we summarize this paper in Section 5.

2. Convolutional factor analysis model

2.1. Brief review of the FA model

FA is a method for modeling the correlations in multidimensional data by using a low-dimensional latent variable. Let $\mathbf{x}_n^{(c,m)}$ represent the n th P -dimensional HRRP sample in the m th aspect-frame of the target c , where $n \in \{1, \dots, N\}$ with N denoting the number of HRRP samples in a frame, $m \in \{1, \dots, M_c\}$ with M_c denoting the number of aspect-frames of the c th target, and $c \in \{1, \dots, C\}$ with C denoting the number of targets. The FA model assumes that $\mathbf{x}_n^{(c,m)}$ is generated via linear mapping from a low-dimensional latent variable plus observation noise and can be mathematically expressed as

$$\mathbf{x}_n^{(c,m)} = \mathbf{D}^{(c,m)} \mathbf{w}_n^{(c,m)} + \boldsymbol{\mu}^{(c,m)} + \boldsymbol{\varepsilon}_n^{(c,m)}, \quad (3)$$

where $\boldsymbol{\mu}^{(c,m)} \in \mathbb{R}^P$ is the mean vector of the m th frame of the c th target; $\mathbf{w}_n^{(c,m)} \in \mathbb{R}^V$ is a latent variable and assumed to be Gaussian distributed as $\mathbf{w}_n^{(c,m)} \sim N(\mathbf{0}, \mathbf{I}_V)$, with $N(\cdot)$ denoting the Gaussian distribution, $\mathbf{I}_V$ being the $V \times V$ dimensional identity matrix, and V denoting the number of factors (here $V < P$); $\mathbf{D}^{(c,m)} \in \mathbb{R}^{P \times V}$ is the factor loading matrix that maps latent variable $\mathbf{w}_n^{(c,m)}$ to the observed sample $\mathbf{x}_n^{(c,m)}$; the noise variable $\boldsymbol{\varepsilon}_n^{(c,m)} \in \mathbb{R}^P \sim N(\mathbf{0}, \boldsymbol{\Psi}^{(c,m)})$ with $\boldsymbol{\Psi}^{(c,m)}$ being a diagonal matrix and the diagonal elements being different. Then, according to the property of the Gaussian variable, we have $\mathbf{x}_n^{(c,m)} \sim N(\boldsymbol{\mu}^{(c,m)}, \mathbf{D}^{(c,m)} \mathbf{D}^{(c,m)T} + \boldsymbol{\Psi}^{(c,m)})$.

The unknown parameters $\Theta = \{\boldsymbol{\mu}^{(c,m)}, \mathbf{D}^{(c,m)}, \boldsymbol{\Psi}^{(c,m)}\}$ in (3) can be estimated via the iterative expectation-maximization (EM) algorithm [41]. In this paper, we refer to the above model as the EM-FA. EM-FA is very sensitive to the initial values of the unknown parameters, which easily causes the model to converge to the local optimal solution. In [42], a hierarchical FA model based on the VB algorithm, i.e., VB-FA, was developed to mitigate the influence of the initial parameter values. The model can be expressed as

$$\begin{aligned} \mathbf{x}_n^{(c,m)} &= \mathbf{D}^{(c,m)} \mathbf{w}_n^{(c,m)} + \boldsymbol{\mu}^{(c,m)} + \boldsymbol{\varepsilon}_n^{(c,m)}, \\ \mathbf{w}_n^{(c,m)} &\sim N(\mathbf{0}, 1/\alpha^{(c,m)} \mathbf{I}_V), \quad \alpha^{(c,m)} \sim \text{Gamma}(a_0, b_0), \\ \mathbf{d}_v^{(c,m)} &\sim N(\mathbf{0}, 1/P \mathbf{I}_P), \\ \boldsymbol{\varepsilon}_n^{(c,m)} &\sim N(\mathbf{0}, 1/\gamma^{(c,m)} \mathbf{I}_P), \quad \gamma^{(c,m)} \sim \text{Gamma}(c_0, d_0), \end{aligned} \quad (4)$$

where $\mathbf{d}_v^{(c,m)}$ is the v th column of the factor loading matrix $\mathbf{D}^{(c,m)}$ with $v \in \{1, \dots, V\}$; $\alpha^{(c,m)}$ and $\gamma^{(c,m)}$ are the precisions of $\mathbf{w}_n^{(c,m)}$

and $\boldsymbol{\varepsilon}_n^{(c,m)}$, respectively; a_0, b_0, c_0 , and d_0 are preset hyperparameters; $\text{Gamma}(\cdot)$ denotes the Gamma distribution parameterized by the arguments in the parentheses.

In this paper, we mainly compare our model with the VB-FA and refer to the VB-FA as the traditional FA model. The traditional FA model can be regarded as a Bayesian form of dictionary learning. For convenience, we call the latent variable $\mathbf{w}_n^{(c,m)}$ as the weight vector and the factor loading matrix $\mathbf{D}^{(c,m)}$ as the dictionary, with its each column being called a dictionary atom.

A statistical model shown in (4) is developed for each aspect-frame because of the target-aspect sensitivity. Therefore, $\sum_{c=1}^C M_c$ independent traditional FA models are required to be learned with the entire training dataset from the C targets.

2.2. CFA model construction

As discussed in Section 1, the convolutional networks are widely used for data representation, especially in the image processing field. Compared to the fully connected networks [43,44], the convolutional networks can extract local features and have fewer parameters to be estimated due to the introduction of convolution kernels [1–8]. By referring to the ideas of convolutional networks, we could develop a convolutional FA model, i.e., CFA model, in which the dictionary $\mathbf{D}^{(c,m)}$ is used as a set of small-scale convolution kernels rather than as a projection matrix.

In our CFA model, the observed sample $\mathbf{x}_n^{(c,m)}$ can be represented as:

$$\mathbf{x}_n^{(c,m)} = \sum_{k=1}^K \mathbf{d}_k^{(c,m)} * \mathbf{w}_{nk}^{(c,m)} + \boldsymbol{\mu}^{(c,m)} + \boldsymbol{\varepsilon}_n^{(c,m)}, \quad (5)$$

where $*$ is the convolution operator; $\boldsymbol{\mu}^{(c,m)} \in \mathbb{R}^P$ is the mean variable of the m th frame for target c and can be directly expressed as $\boldsymbol{\mu}^{(c,m)} = \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n^{(c,m)}$; $\boldsymbol{\varepsilon}_n^{(c,m)} \in \mathbb{R}^P$ represents the noise variable; $\mathbf{d}_k^{(c,m)} \in \mathbb{R}^J$, with $J \ll P$, is the k th dictionary atom of dictionary $\mathbf{D}^{(c,m)}$, and $\mathbf{D}^{(c,m)} = \{\mathbf{d}_k^{(c,m)}\}_{k=1, \dots, K}$ with K being the number of dictionary atoms (usually $K < P$); $\mathbf{w}_{nk}^{(c,m)} \in \mathbb{R}^L$ represents the weight vector of the dictionary atom $\mathbf{d}_k^{(c,m)}$ for observed data $\mathbf{x}_n^{(c,m)}$, and the dimension of $\mathbf{w}_{nk}^{(c,m)}$, i.e., L , is equal to $P - J + 1$. Let $\{\mathbf{w}_{nki}^{(c,m)}\}_{i=1, \dots, L}$ represent the elements of $\mathbf{w}_{nk}^{(c,m)}$. Since the size of all weight vectors $\{\mathbf{w}_{nk}^{(c,m)}\}_{k=1, \dots, K}$, i.e., $(P - J + 1) \times K$, is much larger than that of input $\mathbf{x}_n^{(c,m)}$, i.e., $P \times 1$, our model is over-complete and encounters the risk of having infinite solutions for the dictionary atoms, where some of them cannot reflect the inputs' structures. Therefore, in this model, we perform max-pooling on each weight vector $\mathbf{w}_{nk}^{(c,m)}$ ($k = 1, \dots, K$), where all elements of $\mathbf{w}_{nk}^{(c,m)}$ are set to zero other than the maximum one in the pooling region to facilitate the learning of the dictionary atoms with certain structures. A similar method was adopted in [1,3].

The construction in (5) may be viewed as a special class of FA models. Specially, we can rewrite (5) as:

$$\mathbf{x}_n^{(c,m)} = \sum_{k=1}^K \sum_{i=1}^L \mathbf{w}_{nki}^{(c,m)} \mathbf{d}_{ki}^{(c,m)} + \boldsymbol{\mu}^{(c,m)} + \boldsymbol{\varepsilon}_n^{(c,m)}, \quad (6)$$

where $\mathbf{d}_{ki}^{(c,m)} \in \mathbb{R}^P$ represents a shifted version of $\mathbf{d}_k^{(c,m)}$. In detail, $\mathbf{d}_k^{(c,m)}$ is firstly padded with zeros to length P , and then $\mathbf{d}_{ki}^{(c,m)}$ is generated via circularly shifting the elements in the zero-padded vector by $i - 1$. The model can be implemented via the following hierarchical Bayesian framework:

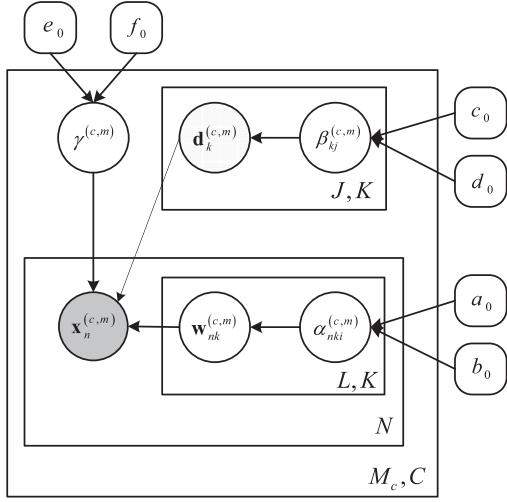


Fig. 2. Graphical representation of the CFA model. The detailed generative process is described in (5) and (7). Here, i and j are the element indexes of the weight vector and dictionary atom, respectively, k is the dictionary atom index, n is the sample index in a frame, m is the frame index, c is the target index, and $\mathbf{x}_n^{(c,m)}$ is the n th HRRP sample in the m th frame of target c . Symbols L, J, K, N, M_c , and C at the edge of a box indicate that there is a stack of such boxes for the indexes i, j, k, n, m and c , where L denotes the dimension of a weight vector, J denotes the dimension of a dictionary atom, K denotes the number of dictionary atoms, N denotes the number of samples in a frame, M_c denotes the number of frames for the target c , and C denotes the number of targets.

$$\begin{aligned} w_{nk}^{(c,m)} &\sim N(0, 1/\alpha_{nk}^{(c,m)}), \quad \alpha_{nk}^{(c,m)} \sim \text{Gamma}(a_0, b_0), \\ d_{kj}^{(c,m)} &\sim N(0, 1/\beta_{kj}^{(c,m)}), \quad \beta_{kj}^{(c,m)} \sim \text{Gamma}(c_0, d_0), \quad j = 1, \dots, J, \\ \mathbf{e}_n^{(c,m)} &\sim N(0, \gamma^{(c,m)^{-1}} \mathbf{I}_p), \quad \gamma^{(c,m)} \sim \text{Gamma}(e_0, f_0). \end{aligned} \quad (7)$$

where J denotes the number of elements in the dictionary atom $\mathbf{d}_k^{(c,m)}$, $d_{kj}^{(c,m)}$ is the j th element of $\mathbf{d}_k^{(c,m)}$, and a_0, b_0, c_0, d_0, e_0 , and f_0 are preset hyperparameters. In our experiments, we set $a_0 = c_0 = 1$, $e_0 = f_0 = 10^{-6}$, $b_0 = 10^{-3}$, $d_0 = 10$, and there is no tuning for these hyperparameters. For further illustration, we present the graphical representation of the CFA model in Fig. 2.

Compared to the traditional FA model, the main differences in model structure originate from three aspects as follows:

First, the convolution operator is introduced into the CFA model, where each dictionary atom $\mathbf{d}_k^{(c,m)}$ is a convolution kernel rather than a basis vector $\mathbf{d}_v^{(c,m)}$ in the traditional FA model. As a convolution kernel, $\mathbf{d}_k^{(c,m)}$ not only has an extremely small size, i.e., $J \ll P$, but is also prone to capture the basic structures in $\mathbf{x}_n^{(c,m)}$, leading to the result that the CFA model learns a fewer dictionary atoms than the traditional FA model, i.e., $K < V$. Consequently, the CFA model offers the potential to more accurately estimate the unknown parameters with limited training data compared with the traditional FA model.

Second, each dictionary atom $\mathbf{d}_k^{(c,m)}$ in the CFA model corresponds to a weight vector $\mathbf{w}_{nk}^{(c,m)}$, instead of a numerical value $w_{nv}^{(c,m)}$, where $w_{nv}^{(c,m)}$ represents the weight of $\mathbf{d}_v^{(c,m)}$ in (4). The weight vector $\mathbf{w}_{nk}^{(c,m)}$ can not only represent the importance of the structural feature extracted by the $\mathbf{d}_k^{(c,m)}$, but also reflect the location of this feature in observed data.

Third, rather than using the same prior for each $\mathbf{d}_v^{(c,m)}$ or $\mathbf{w}_n^{(c,m)}$ in (4), our CFA model assigns different priors to each element in $\mathbf{d}_k^{(c,m)}$ or $\mathbf{w}_{nk}^{(c,m)}$, thus providing more flexibility in the learning of observations' structural information.

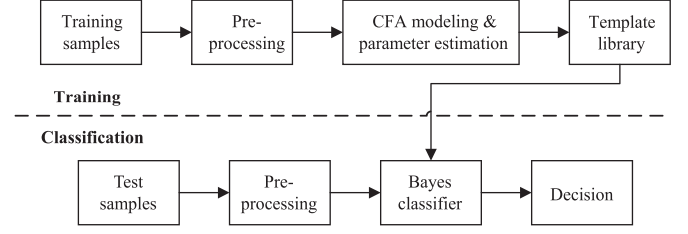


Fig. 3. Flowchart of the radar HRRP statistical recognition based on the CFA model.

3. Recognition system based on the CFA model

In this paper, we build a CFA-based recognition scheme for HRRP data, which is depicted in Fig. 3. The integrated scheme includes two stages, i.e., training stage and classification stage.

In the training stage, raw HRRP data firstly need to be pre-processed to deal with several issues, such as target-aspect, time-shift, and amplitude-scale sensitivities, which will be discussed in detail in Section 4.2. Next, a CFA model is built for each frame of each target, and subsequently, the model parameter is estimated via variational Bayesian (VB) algorithm, i.e., model learning. Finally, all learned model parameters are stored as templates for Bayes classifier construction in the classification phase. In the classification stage, after data preprocessing for a test HRRP sample, the Bayes classifier is constructed, with which the class membership of a test sample is determined based on the frame-conditional likelihoods of all frames from all classes. In detail, the frame-conditional likelihoods of a test sample given each target can be estimated according to the stored templates, and then, the maximum one in each target is selected as the corresponding class-conditional likelihood. Lastly, this test sample is assigned to the class with the maximum class-conditional likelihood.

In this section, we provide the posterior inference of model parameters and introduce an estimation method for the frame-conditional likelihood based on the CFA model.

3.1. Posterior inference exploiting the convolution property

In the training stage, we seek to estimate the posterior distribution of model parameters. According to Bayes' rule, the posterior probability density function (pdf) of the model parameter $\Phi^{(c,m)}$ in the m th aspect-frame of the target c , given the observation data $\mathbf{X}^{(c,m)}$ and the hyperparameters $\theta^{(c,m)}$, can be expressed as

$$\begin{aligned} p(\Phi^{(c,m)} | \mathbf{X}^{(c,m)}, \theta^{(c,m)}) &= \frac{p(\mathbf{X}^{(c,m)} | \Phi^{(c,m)}, \theta^{(c,m)}) p(\Phi^{(c,m)} | \theta^{(c,m)})}{\int p(\mathbf{X}^{(c,m)} | \Phi^{(c,m)}, \theta^{(c,m)}) p(\Phi^{(c,m)} | \theta^{(c,m)}) d\Phi^{(c,m)}}, \end{aligned} \quad (8)$$

where the denominator $\int p(\mathbf{X}^{(c,m)} | \Phi^{(c,m)}, \theta^{(c,m)}) p(\Phi^{(c,m)} | \theta^{(c,m)}) d\Phi^{(c,m)}$ is the marginal likelihood (model evidence). In the proposed hierarchical model, the density function at one layer is conjugated to the density function at the layer above it, and therefore, inference can readily be implemented by Gibbs sampling with Markov Chain Monte Carlo (MCMC) [37] and VB [38,39] methods. Here, we employ VB inference as a compromise between accuracy and computational cost for large-scale problems.

VB inference provides a computationally tractable way which seeks a distribution $q(\Phi^{(c,m)})$ to approximate the exact posterior of the latent variables $p(\Phi^{(c,m)} | \mathbf{X}^{(c,m)}, \theta^{(c,m)})$. Consider the marginal log likelihood of the observations:

$$\ln p(\mathbf{X}^{(c,m)} | \boldsymbol{\theta}^{(c,m)}) = L(q(\boldsymbol{\Phi}^{(c,m)})) + KL(q(\boldsymbol{\Phi}^{(c,m)}) \parallel p(\boldsymbol{\Phi}^{(c,m)} | \mathbf{X}^{(c,m)}, \boldsymbol{\theta}^{(c,m)})), \quad (9)$$

where $L(q(\boldsymbol{\Phi}^{(c,m)})) = \int q(\boldsymbol{\Phi}^{(c,m)}) \ln [p(\mathbf{X}^{(c,m)} | \boldsymbol{\Phi}^{(c,m)}, \boldsymbol{\theta}^{(c,m)}) / q(\boldsymbol{\Phi}^{(c,m)})] d\boldsymbol{\Phi}^{(c,m)}$, and $KL(q(\boldsymbol{\Phi}^{(c,m)}) \parallel p(\boldsymbol{\Phi}^{(c,m)} | \mathbf{X}^{(c,m)}, \boldsymbol{\theta}^{(c,m)})) = \int q(\boldsymbol{\Phi}^{(c,m)}) \ln [q(\boldsymbol{\Phi}^{(c,m)}) / p(\boldsymbol{\Phi}^{(c,m)} | \mathbf{X}^{(c,m)}, \boldsymbol{\theta}^{(c,m)})] d\boldsymbol{\Phi}^{(c,m)}$ is the Kullback–Leibler (KL) divergence between the variational distribution $q(\boldsymbol{\Phi}^{(c,m)})$ and the exact posterior $p(\boldsymbol{\Phi}^{(c,m)} | \mathbf{X}^{(c,m)}, \boldsymbol{\theta}^{(c,m)})$. $L(q(\boldsymbol{\Phi}^{(c,m)}))$ is a lower bound for $\ln p(\mathbf{X}^{(c,m)} | \boldsymbol{\theta}^{(c,m)})$ because of $KL(q(\boldsymbol{\Phi}^{(c,m)}) \parallel p(\boldsymbol{\Phi}^{(c,m)} | \mathbf{X}^{(c,m)}, \boldsymbol{\theta}^{(c,m)}))$ being nonnegative. Therefore, the KL divergence between $q(\boldsymbol{\Phi}^{(c,m)})$ and $p(\boldsymbol{\Phi}^{(c,m)} | \mathbf{X}^{(c,m)}, \boldsymbol{\theta}^{(c,m)})$ is minimized, i.e., the similarity between the variational distribution and the true posterior is maximized, when the lower bound is maximized.

To make the computation of lower bound $L(q(\boldsymbol{\Phi}^{(c,m)}))$ tractable, we need to assume that $q(\boldsymbol{\Phi}^{(c,m)})$ has a factorized form $q(\boldsymbol{\Phi}^{(c,m)}) = \prod_k q_k(\boldsymbol{\Phi}^{(c,m)})$, where $q_k(\boldsymbol{\Phi}^{(c,m)})$ represents each latent variable. This approximation greatly simplifies the derivation since all variational posteriors are now in the same form as the conjugate. With an appropriate choice, $q_k(\boldsymbol{\Phi}^{(c,m)})$ can be iteratively updated so that $L(q(\boldsymbol{\Phi}^{(c,m)}))$ approaches to $\ln p(\mathbf{X}^{(c,m)} | \boldsymbol{\theta}^{(c,m)})$ until convergence.

In our model, the observed data $\mathbf{X}^{(c,m)} = \{\mathbf{x}_n^{(c,m)}\}_{n=1}^N$, latent variables $\boldsymbol{\Phi}^{(c,m)} = \{\mathbf{w}_{nk}^{(c,m)}, \mathbf{d}_k^{(c,m)}, \alpha_{nki}^{(c,m)}, \beta_{kj}^{(c,m)}, \gamma^{(c,m)}\}_{i=1, j=1, k=1, n=1}^{L, J, K, N}$, and hyperparameters $\boldsymbol{\theta}^{(c,m)} = \{a_0, b_0, c_0, d_0, e_0, f_0\}$. The VB updated equations for the posterior distributions of $\boldsymbol{\Phi}^{(c,m)}$ are expressed as follows:

- 1) For $\alpha_{nki}^{(c,m)}$, we have $q(\alpha_{nki}^{(c,m)}) = \text{Gamma}(\alpha_{nki}^{(c,m)} | \tilde{a}_{nki}^{(c,m)}, \tilde{b}_{nki}^{(c,m)})$, and the updated equations for $q(\alpha_{nki}^{(c,m)})$ are

$$\begin{aligned} \tilde{a}_{nki}^{(c,m)} &= a_0 + \frac{1}{2} \\ \tilde{b}_{nki}^{(c,m)} &= b_0 + \frac{1}{2} \langle (\mathbf{w}_{nki}^{(c,m)})^2 \rangle, \end{aligned} \quad (10)$$

where $\langle \cdot \rangle$ indicates the posterior expectation.

- 2) For $\mathbf{w}_{nk}^{(c,m)}$, we have $q(\mathbf{w}_{nk}^{(c,m)}) = N(\mathbf{w}_{nk}^{(c,m)} | \mathcal{S}_{nki}^{(c,m)}, \Sigma_{nki}^{(c,m)})$, with

$$\begin{aligned} \Sigma_{nki}^{(c,m)} &= 1 / (\langle \alpha_{nki}^{(c,m)} \rangle + \langle \gamma^{(c,m)} \rangle \langle \|\mathbf{d}_k^{(c,m)}\|_2^2 \rangle) \\ \mathcal{S}_{nki}^{(c,m)} &= \Sigma_{nki}^{(c,m)} \langle \gamma^{(c,m)} \rangle [\mathbf{x}_{-n}^{(c,m)T} \langle \mathbf{d}_k^{(c,m)} \rangle + \langle \mathbf{w}_{nki}^{(c,m)} \rangle \langle \|\mathbf{d}_k^{(c,m)}\|_2^2 \rangle]. \end{aligned} \quad (11)$$

Here $\mathbf{x}_{-n}^{(c,m)} = \mathbf{x}_n^{(c,m)} - \boldsymbol{\mu}^{(c,m)} - \sum_{k=1}^K \mathbf{d}_k^{(c,m)} * \mathbf{w}_{nk}^{(c,m)}$. By taking advantage of the convolution property, we simultaneously update the posterior mean and variance of the weights for all the shifted versions of one dictionary atom. Consequently,

$$\begin{aligned} \Sigma_{nk}^{(c,m)} &= \mathbf{1}_L \otimes (\langle \alpha_{nk}^{(c,m)} \rangle + \langle \gamma^{(c,m)} \rangle \langle \|\mathbf{d}_k^{(c,m)}\|_2^2 \rangle \mathbf{1}_L) \\ \mathcal{S}_{nk}^{(c,m)} &= \Sigma_{nk}^{(c,m)} \odot \langle \gamma^{(c,m)} \rangle \\ &\quad \left(\langle \mathbf{x}_{-n}^{(c,m)} \rangle * \langle \mathbf{d}_k^{(c,m)} \rangle + \langle \mathbf{w}_{nk}^{(c,m)} \rangle \langle \|\mathbf{d}_k^{(c,m)}\|_2^2 \rangle \right), \end{aligned} \quad (12)$$

where $\mathbf{1}_L$ represents an L -dimensional vector with each element being 1; $\alpha_{nk}^{(c,m)} = [\alpha_{nkl}^{(c,m)}, \dots, \alpha_{nkl}^{(c,m)}]^T$, and $\mathbf{d}_k^{(c,m)} \in \mathbb{R}^J$ represents the vertically flipped version of the $\mathbf{d}_k^{(c,m)}$; the symbol $\odot$ is the element-wise product operator and $\otimes$ is the element-wise division operator.

After updating $\mathbf{w}_{nk}^{(c,m)}$, we perform max-pooling on it at each iteration. In detail, within each max-pooling region, we set all coefficients to zero within each max-pooling region, except for the maximum-amplitude coefficient.

- 3) For $\beta_{kj}^{(c,m)}$, we have $q(\beta_{kj}^{(c,m)}) = \text{Gamma}(\beta_{kj}^{(c,m)} | \tilde{c}_{kj}^{(c,m)}, \tilde{d}_{kj}^{(c,m)})$, and the updated equations for $q(\beta_{kj}^{(c,m)})$ are expressed as

$$\begin{aligned} \tilde{c}_{kj}^{(c,m)} &= c_0 + \frac{1}{2} \\ \tilde{d}_{kj}^{(c,m)} &= d_0 + \frac{1}{2} \langle (d_{kj}^{(c,m)})^2 \rangle. \end{aligned} \quad (13)$$

- 4) For $\mathbf{d}_k^{(c,m)}$, we have $q(\mathbf{d}_k^{(c,m)}) = N(\mathbf{d}_k^{(c,m)} | \xi_{kj}^{(c,m)}, \Lambda_{kj}^{(c,m)})$, where $\xi_{kj}^{(c,m)}$ and $\Lambda_{kj}^{(c,m)}$ are the j th elements of the following vectors $\xi_k^{(c,m)}$ and $\Lambda_k^{(c,m)}$, respectively.

$$\begin{aligned} \Lambda_k^{(c,m)} &= \mathbf{1}_J \otimes \left(\langle \beta_k^{(c,m)} \rangle + \langle \gamma^{(c,m)} \rangle \sum_{n=1}^N \langle \|\mathbf{w}_{nk}^{(c,m)}\|_2^2 \rangle \mathbf{1}_J \right) \\ \xi_k^{(c,m)} &= \Lambda_k^{(c,m)} \odot \langle \gamma^{(c,m)} \rangle \sum_{n=1}^N \left(\langle \mathbf{x}_{-n}^{(c,m)} \rangle * \langle \mathbf{w}_{nk}^{(c,m)} \rangle \right. \\ &\quad \left. + \langle \mathbf{d}_k^{(c,m)} \rangle \langle \|\mathbf{w}_{nk}^{(c,m)}\|_2^2 \rangle \right), \end{aligned} \quad (14)$$

where $\mathbf{1}_J$ represents a J -dimensional vector with each element being 1; $\beta_k^{(c,m)} = [\beta_{k1}^{(c,m)}, \dots, \beta_{kJ}^{(c,m)}]^T$, and $\mathbf{w}_{nk}^{(c,m)} \in \mathbb{R}^L$ represents the vertically flipped version of $\mathbf{w}_{nk}^{(c,m)}$.

Usually, in order to estimate $q(\mathbf{d}_k^{(c,m)})$, we first need to write out the expected complete log likelihood with respect to all other variational posteriors, i.e., $\langle \ln p(\mathbf{X}^{(c,m)}, \boldsymbol{\Phi}^{(c,m)} | \boldsymbol{\theta}^{(c,m)}) \rangle_{q(\mathbf{w}_{nk}^{(c,m)}), q(\alpha_{nki}^{(c,m)}), q(\beta_{kj}^{(c,m)}), q(\gamma^{(c,m)})}$, and then neglect terms not depending on the $\mathbf{d}_k^{(c,m)}$. After the complicated deformation and arrangement of the expected complete log likelihood, we can finally identify the posterior of $\mathbf{d}_k^{(c,m)}$. However, on the basis of the commutative law of the convolution operation, we can directly derive (14) by replacing $\mathbf{d}_k^{(c,m)}$ in (12) with $\mathbf{w}_{nk}^{(c,m)}$, replacing $\alpha_{nk}^{(c,m)}$ in (12) with $\beta_k^{(c,m)}$, and summing over n (because the dictionary atom $\mathbf{d}_k^{(c,m)}$ is shared for all $\{\mathbf{x}_n^{(c,m)}\}_{n=1, \dots, N}$). This technique can help to avoid the complex derivation mentioned above.

- 5) For $\gamma^{(c,m)}$, we have $q(\gamma^{(c,m)}) = \text{Gamma}(\gamma^{(c,m)} | \tilde{e}^{(c,m)}, \tilde{f}^{(c,m)})$, and the updated equations for $q(\gamma^{(c,m)})$ are

$$\begin{aligned} \tilde{e}^{(c,m)} &= e_0^{(c,m)} + \frac{NP}{2} \\ \tilde{f}^{(c,m)} &= f_0^{(c,m)} + \frac{1}{2} \sum_{n=1}^N \langle \|\mathbf{x}_n^{(c,m)} - \sum_{k=1}^K \langle \mathbf{d}_k^{(c,m)} \rangle * \langle \mathbf{w}_{nk}^{(c,m)} \rangle\|_2^2 + \mathbf{v}^{(c,m)} \rangle, \end{aligned} \quad (15)$$

where $\mathbf{v}^{(c,m)} = \sum_{k=1}^K (\langle \|\mathbf{d}_k^{(c,m)} * \mathbf{w}_{nk}^{(c,m)}\|_2^2 \rangle - \langle \mathbf{d}_k^{(c,m)} \rangle * \langle \mathbf{w}_{nk}^{(c,m)} \rangle \langle \|\mathbf{d}_k^{(c,m)}\|_2^2 \rangle)$.

In the following experiments, we terminate the algorithm when the change of $L(q(\boldsymbol{\Phi}))$ is negligibly small (e.g., the lower bound variation rate between two iterations is lower than 10^{-5}).

3.2. Classification via Bayes classifier

The Bayes classifier is used in predicting the label of a test sample, with which the class membership of that test sample is determined based on the target index of the maximum frame-conditional likelihood among all aspect-frames of all targets. In this paper, we adopt a method similar to that in [40] to calculate the frame-conditional likelihood.

For the n th sample in the m th frame of the target c , we first rewrite the CFA model, i.e., (5), to make it consistent with the traditional FA model:

$$\mathbf{x}_n^{(c,m)} = \widehat{\mathbf{D}}^{(c,m)} \widehat{\mathbf{w}}_n^{(c,m)} + \boldsymbol{\mu}^{(c,m)} + \boldsymbol{\epsilon}_n^{(c,m)}, \quad (16)$$

where $\hat{\mathbf{D}}^{(c,m)} = [\mathbf{d}_{11}^{(c,m)}, \dots, \mathbf{d}_{1L}^{(c,m)}, \dots, \mathbf{d}_{K1}^{(c,m)}, \dots, \mathbf{d}_{KL}^{(c,m)}] \in \mathbb{R}^{P \times KL}$ with $\mathbf{d}_{KL}^{(c,m)}$ representing the L th shifted version of the dictionary atom $\mathbf{d}_K^{(c,m)}$ and other notations in $\hat{\mathbf{D}}^{(c,m)}$ being similar; $\hat{\mathbf{W}}_n^{(c,m)} = [w_{n11}^{(c,m)}, \dots, w_{n1L}^{(c,m)}, \dots, w_{nK1}^{(c,m)}, \dots, w_{nKL}^{(c,m)}]^T \in \mathbb{R}^{KL}$ with $w_{nKL}^{(c,m)}$ being the weight of the $\mathbf{d}_{KL}^{(c,m)}$ and similar meaning for other notations.

After learning the models for all frames of all targets, $q(\Phi^{(c,m)})$ is obtained, and then $q(\hat{\mathbf{D}}^{(c,m)})$ and $q(\hat{\mathbf{W}}_n^{(c,m)})$ can also be easily obtained. For a test sample $\mathbf{x}^*$, the frame-conditional likelihood given the frame m of the target c can be estimated as follows:

$$p(\mathbf{x}^*|m, c) = \int N(\mathbf{x}^* | \langle \hat{\mathbf{D}}^{(c,m)} \rangle \hat{\mathbf{W}}^{(c,m)} + \boldsymbol{\mu}^{(c,m)}, \langle \gamma^{(c,m)} \rangle^{-1} \mathbf{I}_P) q(\hat{\mathbf{W}}^{(c,m)}) d\hat{\mathbf{W}}^{(c,m)}, \quad (17)$$

where $\langle \hat{\mathbf{D}}^{(c,m)} \rangle$ and $\langle \gamma^{(c,m)} \rangle$ are the posterior means of $\hat{\mathbf{D}}^{(c,m)}$ and $\gamma^{(c,m)}$, respectively. $\hat{\mathbf{W}}^{(c,m)}$ is the statistical weight of the m th frame for target c , and $q(\hat{\mathbf{W}}^{(c,m)})$ denotes the posterior pdf for $\hat{\mathbf{W}}^{(c,m)}$, which can be simplified as

$$\begin{aligned} q(\hat{\mathbf{W}}^{(c,m)}) &= N(\hat{\mathbf{W}}^{(c,m)} | \eta^{(c,m)}, \boldsymbol{\Omega}^{(c,m)}) \\ \eta^{(c,m)} &= \frac{1}{N} \sum_{n=1}^N \langle \hat{\mathbf{W}}_n^{(c,m)} \rangle \\ \boldsymbol{\Omega}^{(c,m)} &= \frac{1}{N} \sum_{n=1}^N (\langle \hat{\mathbf{W}}_n^{(c,m)} \rangle - \eta^{(c,m)}) (\langle \hat{\mathbf{W}}_n^{(c,m)} \rangle - \eta^{(c,m)})^T, \end{aligned} \quad (18)$$

where $\langle \hat{\mathbf{W}}_n^{(c,m)} \rangle$ represents the posterior mean of $\hat{\mathbf{W}}_n^{(c,m)}$. Ref. [40] used a similar estimator from Gibbs sampling for the compressive sensing application.

Then, we can calculate the integrals in (17) and obtain the frame-conditional likelihood of $\mathbf{x}^*$ given the frame m of the target c as follows:

$$\begin{aligned} p(\mathbf{x}^*|m, c) &= N(\mathbf{x}^* | \chi^{(c,m)}, \boldsymbol{\Gamma}^{(c,m)}) \\ \chi^{(c,m)} &= \boldsymbol{\mu}^{(c,m)} + \langle \hat{\mathbf{D}}^{(c,m)} \rangle \eta^{(c,m)} \\ \boldsymbol{\Gamma}^{(c,m)} &= \langle \hat{\mathbf{D}}^{(c,m)} \rangle \boldsymbol{\Omega}^{(c,m)} \langle \hat{\mathbf{D}}^{(c,m)} \rangle^T + \langle \gamma^{(c,m)} \rangle^{-1} \mathbf{I}_P. \end{aligned} \quad (19)$$

Finally, we can predict the class membership of the test sample $\mathbf{x}^*$ according to the expression (19).

4. Experimental results

4.1. Synthetic data

In this subsection, we design several simulation examples to demonstrate that our CFA model is capable of extracting the basic structures of observed data, and this property can considerably decrease the number of dictionary atoms to be learned. Moreover, the relationship between the dimension of the dictionary atom and the number of dictionary atoms is also investigated by using the synthetic data.

4.1.1. Illustration using simple synthetic data

The building block shown in Fig. 4(a) is used to constitute 100 noise-free samples (we refer to them as the original samples), two of which are shown in Fig. 4(b). Each original sample is generated by using three building blocks and situating them arbitrarily. The sizes of the building block and synthetic samples are 9×1 and 72×1 , respectively. In order to keep the signal-to-noise ratio (SNR) of the synthetic data consistent with that of the measured data (the SNR of measured HRRP data is 22 dB to 30 dB), we add simulated white noises to the original samples. The SNR of each

noisy sample is 22 dB, and the two noisy samples from Fig. 4(b) are shown in Fig. 4(c). The SNR is defined as

$$\text{SNR} = 10 \times \log_{10} \left(\frac{\bar{P}_x}{P_{\text{Noise}}} \right) = 10 \times \log_{10} \left(\frac{\sum_{p=1}^P P_{x_p}}{P \times P_{\text{Noise}}} \right), \quad (20)$$

where $\bar{P}_x$ denotes the average power of the original sample, P_{x_p} denotes the power of the p th dimension in the original sample, P denotes the dimension of the sample, and P_{Noise} denotes the power of noise. Thus, $10^{\frac{\text{SNR}}{10}}$ equals the ratio of the average power of the original sample to the power of the noise.

We model the noisy synthetic data above via the traditional FA model and our CFA model. In this experiment, we determine the number of dictionary atoms, i.e., V for the traditional FA model and K for the CFA model, by performing two steps. In the first step, a candidate range $\mathcal{M}$ for V and K is pre-specified (here, $\mathcal{M} = \{1, \dots, 20\}$). For each $V \in \mathcal{M}$ or $K \in \mathcal{M}$, parameter estimations are obtained by the corresponding VB algorithm. In the second step, an average relative reconstruction error (AR²E) for each $V \in \mathcal{M}$ or $K \in \mathcal{M}$ is calculated, and V_{opt} or K_{opt} corresponding to the minimum AR²E is selected as an optimal estimation for V or K . Here, the AR²E is defined as $\frac{1}{N} \sum_{n=1}^N \frac{\|\hat{\mathbf{x}}_n - \mathbf{x}_n\|_2}{\|\mathbf{x}_n\|_2}$, where $\mathbf{x}_n$ denotes the n th original sample, $\hat{\mathbf{x}}_n$ represents the corresponding recovery sample, and N denotes the number of samples.

Fig. 5(a) and (c) show the dictionary atoms learned with the traditional FA model and the CFA model, with the number of their dictionary atoms being optimal. Fig. 5(b) and (d) depict the two reconstructed samples from Fig. 4(b) that correspond to the traditional FA model and the CFA model, respectively. As shown in Fig. 5(a) and (c), the inferred dictionary atom of our model is similar to the building block in this experiment, i.e., the inverted triangle; while the learned dictionary atoms of the traditional FA model cannot reflect the observations' structural information. Owing to this property, our model only needs to learn one dictionary atom, whereas the traditional FA model learns six dictionary atoms. In addition, the minimum AR²Es of the traditional FA model and the CFA model are 2.57% and 2.03%, respectively, which demonstrate that our method can learn the model better with a fewer dictionary atoms.

4.1.2. Illustration using complex synthetic data

To verify the characteristics of our CFA model under more complicated situations, we further perform an analysis on 200 samples generated by five building blocks. Fig. 6(a) shows the five building blocks, including regular triangle, inverted triangle, horizontal line, and left and right slashes. Two hundred original samples are generated by arbitrarily situating the five building blocks, and two examples are shown in Fig. 6(b). Simulated white noises are added to the original samples, and the SNR of each noisy sample is 22 dB. The two noisy samples from Fig. 6(b) are presented in Fig. 6(c). In addition, the sizes of the building blocks and synthetic samples are 9×1 and 107×1 , respectively.

We also model the noisy synthetic data mentioned above via the traditional FA model and the CFA model. The same method as that in the previous experiment is adopted to determine the optimal values of V and K in this experiment, and the candidate range for the two models is from 1 to 50, i.e., $\mathcal{M} = \{1, \dots, 50\}$. Fig. 7(a) and (c) show the learned dictionary atoms via the traditional FA model and the CFA model, with their dictionary atom number being optimal. The reconstructed samples from Fig. 6(b) corresponding to the two models are depicted in Fig. 7(b) and (d). Notably, the optimal number of dictionary atoms for the traditional FA model is 31, and 12 dictionary atoms with large average relative reconstruction energy (ARRE) are shown in Fig. 7(a) for clarity. The ARRE of

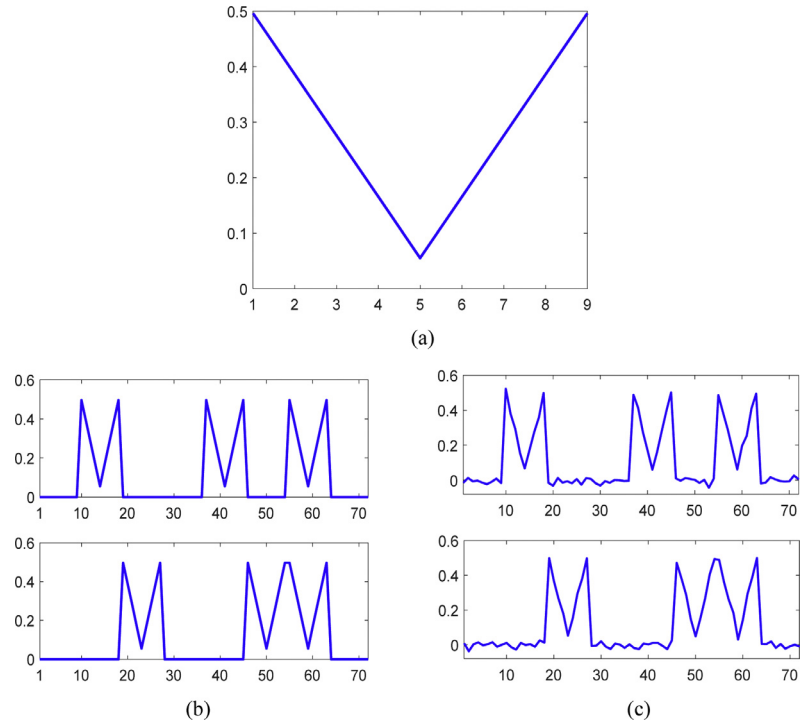


Fig. 4. (a) A vector of size 9×1 is used as a building block for synthetic data generation. (b) Two of 100 original samples are shown here, which are generated by using three building blocks depicted in (a) and situating them arbitrarily. Each sample is of size 72×1 . (c) Simulated white noises are added to the original samples, and two noisy samples are shown here. The SNR of each noisy synthetic sample is 22 dB.

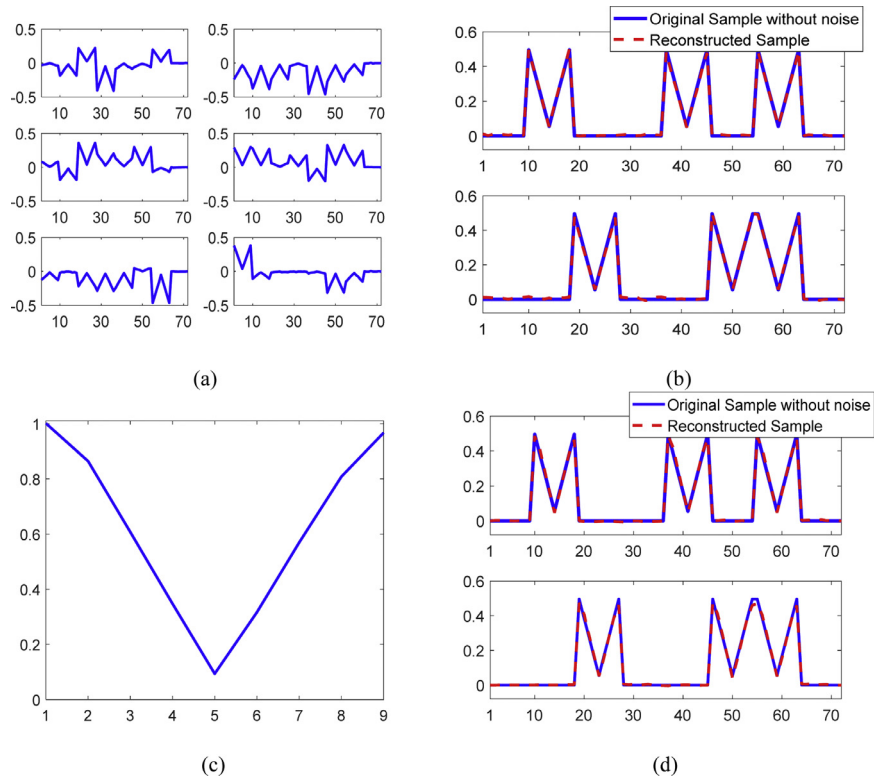


Fig. 5. Learned dictionary atoms and the reconstructed samples from Fig. 4(b) by using the traditional FA model and CFA model, respectively. All experimental results shown here are obtained under the minimum AR^2E condition. (a) and (b) show the learned dictionary atoms and reconstructed samples from the traditional FA model, respectively. (c) and (d) show the inferred dictionary atom and the reconstructed samples via the CFA model, respectively.

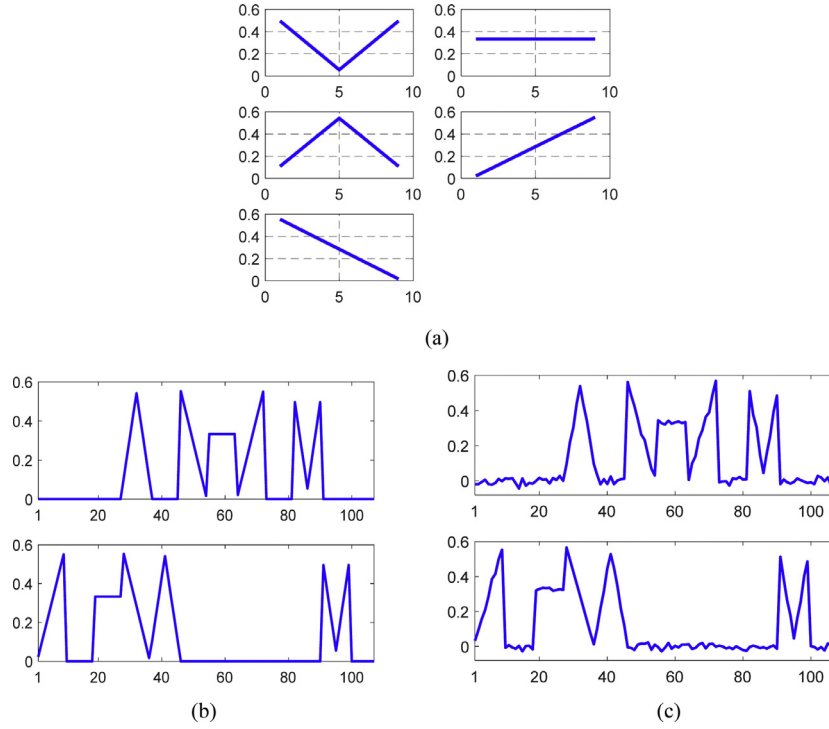


Fig. 6. (a) Five 9×1 vectors are used as building blocks for synthetic data generation. (b) Two hundred original samples are generated by using the five building blocks depicted in (a) and situating them arbitrarily. Two examples of them are shown here, and each sample has a size of 107×1 . (c) Two noisy samples with the SNR of 22 dB are presented here.

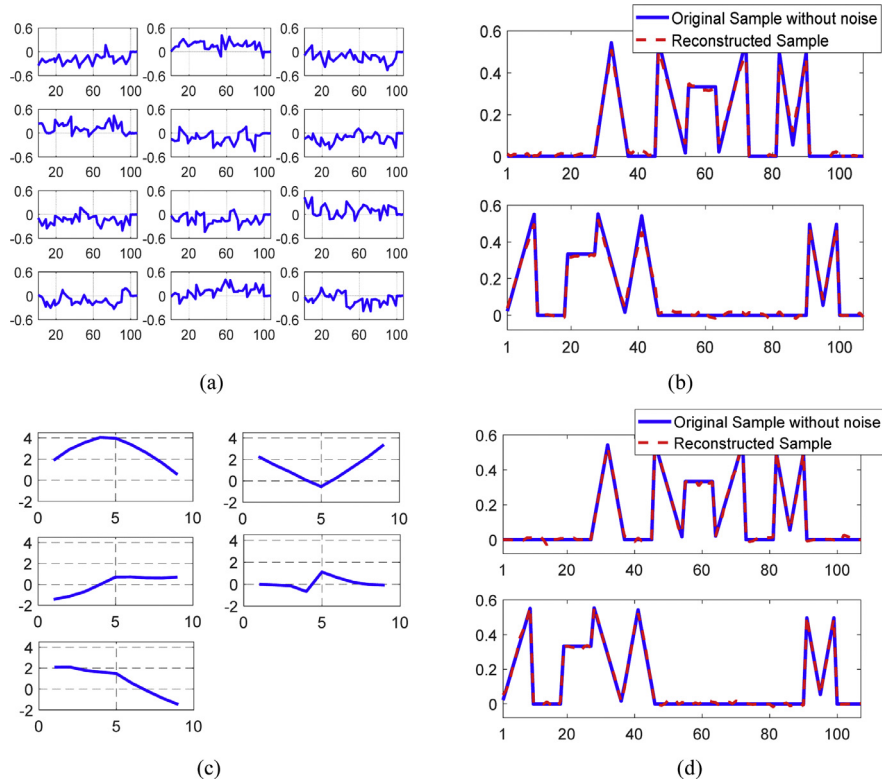


Fig. 7. Learned dictionary atoms and the reconstructed samples from Fig. 6(b) by using the traditional FA model and the CFA model, respectively. All results are obtained under the condition that the two models have minimum reconstruction errors. (a) shows the inferred dictionary atoms that correspond to the traditional FA model, and 12 of them are shown here. (b) shows the two reconstructed samples from Fig. 6(b) via the traditional FA model, and (c) and (d) show the inferred dictionary atoms and the reconstructed samples via the CFA model, respectively.

the v th dictionary atom for the traditional FA model, i.e., $\mathbf{d}_v$, is defined as

$$\frac{1}{N} \sum_{n=1}^N \frac{\|\mathbf{w}_{nv} \mathbf{d}_v\|_2^2}{\|\mathbf{D} \mathbf{w}_n\|_2^2}, \quad (21)$$

where w_{nv} represents the v th element of $\mathbf{w}_n$ and N denotes the number of samples. The ARRE of the k th dictionary atom for the CFA model, i.e., $\mathbf{d}_k$, is defined as

$$\frac{1}{N} \sum_{n=1}^N \frac{\|\mathbf{d}_k * \mathbf{w}_{nk}\|_2^2}{\left\| \sum_{k=1}^K \mathbf{d}_k * \mathbf{w}_{nk} \right\|_2^2}. \quad (22)$$

Comparison of Fig. 7(a) and (c) indicates that the inferred dictionary atoms via the CFA model are similar to the building blocks in Fig. 6(a), and our model needs to learn much fewer dictionary atoms than the traditional FA model. Moreover, the minimum AR²E of the traditional FA model and the CFA model are 0.80% and 0.50%, respectively; therefore, our model is learned better.

As previously discussed, the dictionary atoms in our CFA model can describe the observations' local structures, and this capability reduces the requirement for the number of dictionary atoms. Actually, the lower the dimension of a dictionary atom is, the more basic the local structure is, and the more shared it is in different parts of observed data. Thus, the data can be represented with a small number of dictionary atoms. In the following, experiments on the same 200 noisy samples are conducted to demonstrate the relationship between the dictionary atom dimension and the number of atoms.

We model the noisy synthetic data via our CFA model, with the dimensions of the dictionary atom being 3, 6, 9, and 15, respectively. The learned dictionaries under the minimum AR²E condition are shown in Fig. 8. As we can see from Fig. 8, when the dimension of the dictionary atom is 3, the inferred dictionary presents simple line features, which are highly shared in different parts of observations, and thus, only two dictionary atoms need to be learned. Complicated curve information is extracted when the dimension of the dictionary atom increases to 6, and its sharing is weakened, leading to the result that more (three) dictionary atoms need to be learned. The learned dictionary atoms are similar to the building blocks in Fig. 6(a) when the dimension of the dictionary atom is 9, i.e., equal to that of the actual dictionary atom. Many more abstract structures are extracted when the dimension of the dictionary atom is 15, and therefore, the CFA model needs to learn more dictionary atoms, i.e., eight dictionary atoms, to represent the observed samples. In summary, the lower the dimension of the dictionary atom is, the smaller the number of required dictionary atoms is.

4.2. Measured data

4.2.1. Data description

We then examine the performance of the CFA model on the measured HRRP data. The measured HRRP data used in this paper (also used in [14,15,17,18,21,22,24–27,30–34]) are ISAR data based on C band radar, and the center frequency is 5520 MHz, the signal bandwidth is 400 MHz, the pulse repetition frequency is 400 Hz, the range resolution is 0.375 m, and the wavelength is 0.05 m. The radar targets are three airplanes, including a big yet plane Yark-42, a small yet plane Cessna Citation S/II, and a propeller plane An-26. The parameters of the airplane targets and measurement radar are shown in Table 1, and physical photos of the three airplanes are presented in Fig. 9. In the experiment, an HRRP is the amplitude of a complex vector that is a part of the received radar echo with the target signal being included. The dimension of this complex vector is 256, i.e., 256 range cells ($P = 256$), both the inphase

Table 1

Parameters of the three airplanes and radar in the experiment.

Radar parameters	Center frequency: 5520 MHz Bandwidth: 400 MHz Pulse repetition frequency: 400 MHz		
Planes	Length (m)	Width (m)	Height (m)
Yark-42	36.38	34.88	9.83
Cessna Citation S/II	14.40	15.90	4.57
An-26	23.80	29.20	9.83

and quadrature components in each range cell are 8 bits. In addition, this HRRP dataset is measured under the cooperative condition with the SNR being 22–30 dB.

We present the projections of target trajectories onto the ground plane in Fig. 10, from which the aspect angle of the airplane can be estimated according to its relative position to the radar. As shown in Fig. 10, the measured data are segmented, and each segment contains 25,600 HRRP samples. In order to test the generalization performance of the recognition method, two requirements for selecting the training data and test data should be considered: (a) the training data should cover almost all the target-aspect angles of the test data; (b) the elevation angles of the test data should be different from those of training data. Therefore, data from the second and a part of the fifth segments of Yark-42, the sixth and seventh segments of Cessna Citation S/II, and the fifth and the sixth segments of An-26 are taken as training samples, and other data segments are taken as test samples in our experiments.

Thirteen training datasets are considered to train the CFA model and several other methods. These 13 datasets have data size expressions of 7×135 , 10×135 , 20×135 , 30×135 , 50×135 , 70×135 , 90×135 , 110×135 , 130×135 , 160×135 , 256×135 , 512×135 , 1024×135 , where 7×135 means that 7 HRRP samples are randomly selected from each of the 135 aspect-frames; the same applies to the other data size expressions. To avoid occasional results caused by random selection and obtain more convincing conclusions, the selection work is repeated 10 times, and all the results are averaged over 10 model runs. In all experiments in Section 4.2, the dimension of each dictionary atom in our CFA model is set to 17, i.e., $J = 17$. The number of the dictionary atoms for each aspect-frame, i.e., K , is the same, and is determined by using the approach for the experiments in Section 4.1. First, we pre-specify a candidate range $\mathcal{M}$ for K ($\mathcal{M} = \{1, \dots, 65\}$ in Section 4.2). Second, for each $K \in \mathcal{M}$, we train a CFA model for each frame. Finally, the K_{opt} corresponding to the highest recognition rate is selected as the optimal estimation to K .

4.2.2. Data preprocessing

Data preprocessing is a critical stage in HRRP-based statistical recognition due to the target-aspect sensitivity, time-shift sensitivity and amplitude-scale sensitivity of HRRP. In the following, we will comprehensively analyze the effects of the implemented preprocessing methods on the robustness of radar HRRP statistical recognition with the CFA model.

A. Target-aspect sensitivity

The HRRP distribution varies with the change of the target-aspect angle. Similar to [21], we use the aspect-frame partition method to deal with the target-aspect sensitivity of HRRP. Specifically, we construct a model for each aspect-frame, where HRRPs are collected from a target-aspect sector roughly without the MTRC. The limitation of target-aspect change for avoiding scatterers' MTRC is

$$\Delta\varphi \leq (\Delta\varphi)_{\text{MTRC}} \stackrel{\text{Def}}{=} \frac{c}{2BL}, \quad (23)$$

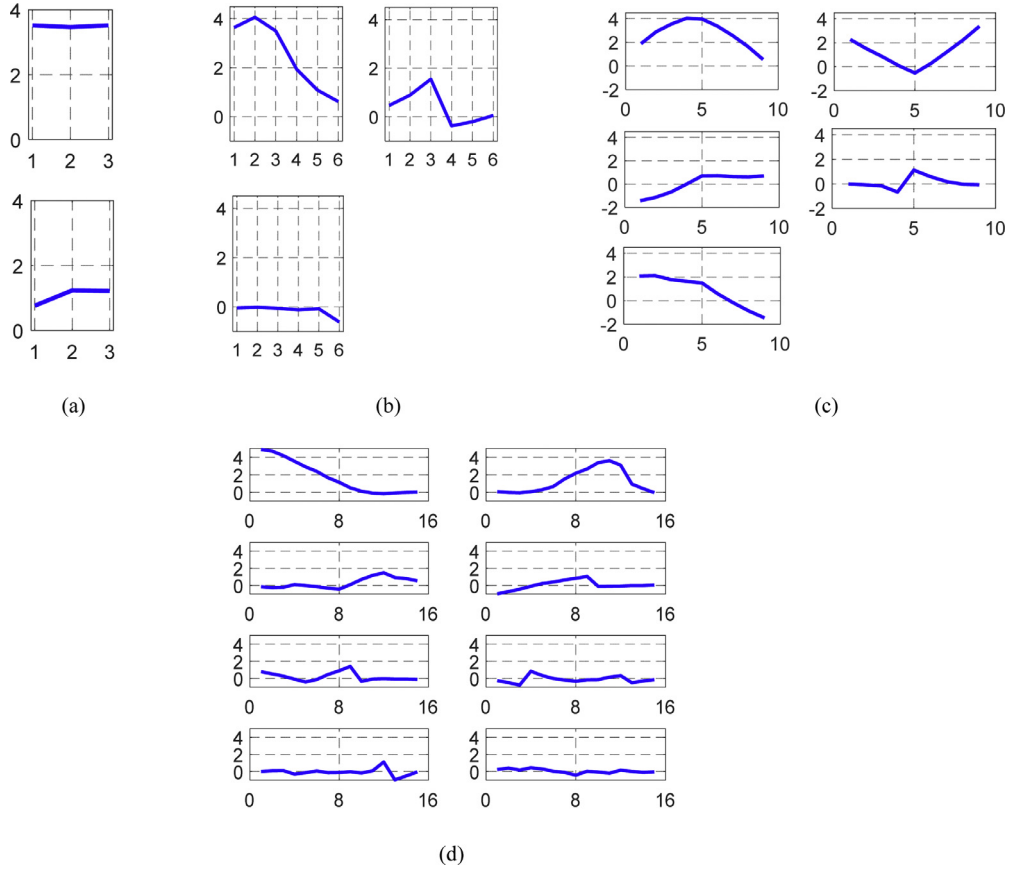


Fig. 8. Learned dictionaries via the CFA model with different dictionary atom dimensions. The dimensions of the dictionary atom in (a), (b), (c), and (d) are 3, 6, 9, and 15, respectively, and all the results shown here are under the minimum AR^2E condition.

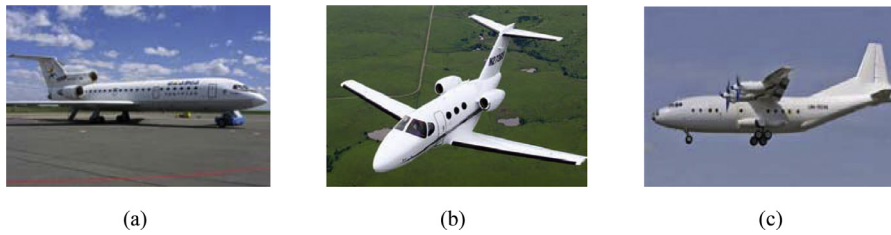


Fig. 9. Physical photos of three airplane targets. (a) Yark-42. (b) Cessna Citation S/II. (c) An-26.

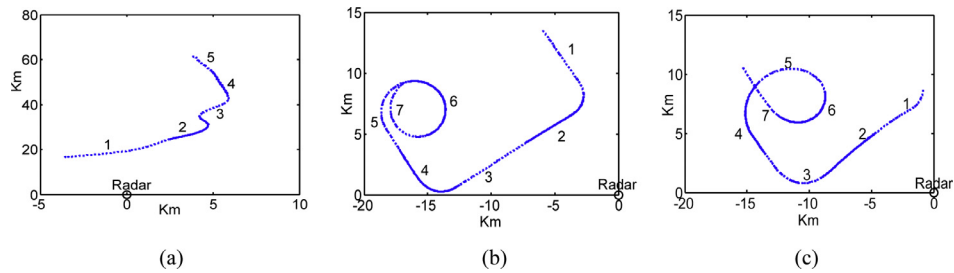


Fig. 10. Projections of target trajectories onto the ground plane. (a) Yark-42. (b) Cessna Citation S/II. (c) An-26.

where B , L , and c are the bandwidth of the radar signal after range compression, the maximum target dimension in cross range, and the speed of light, respectively. In this paper, there are 35 aspect-frames for Yark-42, 50 for Cessna Citation S/II, and 50 for An-26, of which each aspect-frame totally has 1024 samples, i.e., $C = 3$, $M_1 = 35$, $M_2 = M_3 = 50$, $N = 1024$.

To verify the effectiveness of the aspect-frame partition method, we construct a CFA model for each target and for each frame of each target, respectively, and show the recognition accuracies under different training data size conditions in Fig. 11. As we can see from Fig. 11, the CFA model building for each frame of each target has better recognition performance than the CFA model only building for each target, which demonstrates that the aspect-frame par-

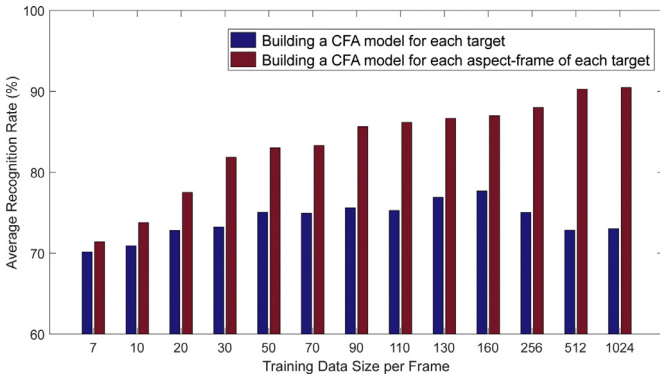


Fig. 11. Variations in recognition performance with training data size via the CFA model building for each target and for each frame of each target. The recognition rates are averaged over 10 runs.

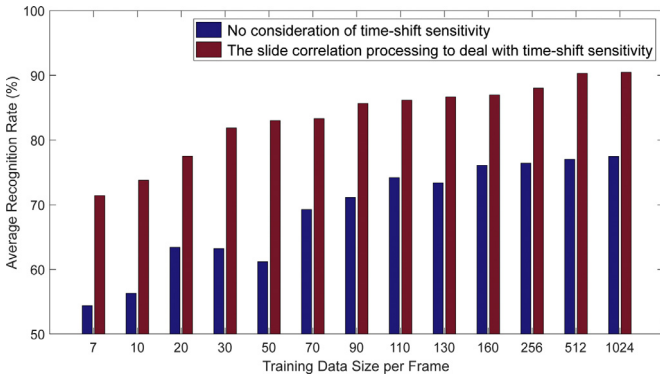


Fig. 12. Variations of recognition performance with training data size, via the CFA models for unaligned and aligned HRRPs. The results are averaged over 10 runs.

tion method can deal with the target-aspect sensitivity of HRRP effectively.

B. Time-shift sensitivity

Since an HRRP is only a part of the received radar echo extracted by a range window where the target signal is included, the position of the target signal in an HRRP can vary with the measurement. This is the time-shift sensitivity of HRRP [26]. In the training stage, a traditional motion compensation method, i.e., slide correlation processing [29], is commonly used for HRRPs from each aspect-frame to deal with the time-shift sensitivity of HRRP training samples. In the classification stage, the slide correlation processing between the test sample and the corresponding frame average profile is usually adopted to compensate the test HRRP sample, when we estimate the frame-conditional likelihood of the test sample given the frame distribution.

In Fig. 12, we show the variations of recognition performance with training data size via the CFA model for unaligned and aligned HRRPs. Fig. 12 indicates that the CFA model for HRRP samples with time-shift compensations can achieve higher recognition rates.

C. Amplitude-scale sensitivity

Since the intensity of an HRRP is a function of radar transmitting power, target distance, radar antenna gain, radar receiver gain, radar system losses, and so on, HRRPs measured by different radars or under different conditions have different amplitude-scales. This is the amplitude-scale sensitivity of HRRP. For radar HRRP statistical recognition, the difference in amplitude-scale among HRRPs greatly affects the likelihood of an HRRP sample given the statistical model, and furthermore, leads to a considerable impact on the

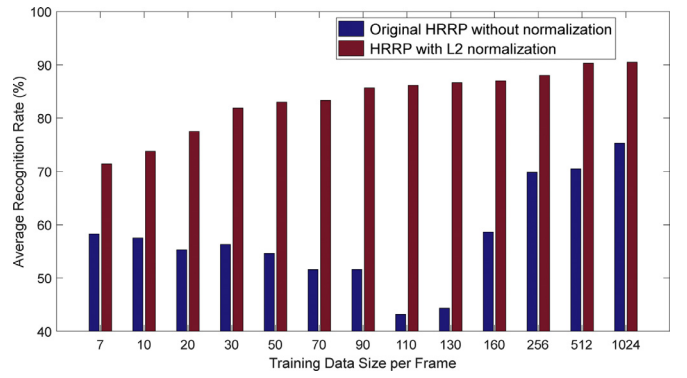


Fig. 13. Recognition performance of the CFA model for original HRRPs without normalizations and L_2 -normalized HRRPs under different training data size conditions. The recognition rates are averaged over 10 runs.

recognition performance of the model. Usually, the amplitude-scale sensitivity of HRRP can be tackled by L_2 normalization [20], i.e., each HRRP sample is divided by its L_2 norm.

To verify the effectiveness of L_2 normalization, we recognize the original HRRPs without normalizations and the L_2 -normalized HRRPs via the CFA model and present the recognition results in Fig. 13. We note that the recognition accuracies of the original HRRPs are much lower than those of the HRRPs with L_2 -normalizations, which demonstrates the effectiveness of L_2 normalization.

4.2.3. Training results analysis

As stated in Section 2, the dictionary learned by our CFA model can reflect the structural information of data. To verify this claim, we train a CFA model using 70 HRRP samples for each frame and select a frame from each airplane target for comparison. Two HRRP training samples from a frame of each target are shown in Fig. 14(a), (c) and (e), and the corresponding learned dictionary atoms are depicted in Fig. 14(b), (d) and (f). Notably, only 128 range cells close to the support for each HRRP sample are shown in Fig. 14(a), (c) and (e) for clarity, and the ARREs of the inferred dictionary atoms for each aspect-frame decrease from left to right and top to bottom.

Comparing Fig. 14(a), (c) and (e), we see that an HRRP sample consists of some different “spikes,” whose number varies greatly with different targets. For HRRPs from Yark-42, there are quite a number of different spikes; while only a few different spikes for Cessna Citation S/II, e.g. two spikes in Fig. 14(c); for An-26, the number of different spikes is larger than that of Cessna Citation S/II and smaller than that of Yark-42. Fig. 14(b) illustrates that most of the learned dictionary atoms of Yark-42 contain multiple spikes. In addition, the ARREs of the six dictionary atoms are 21.62%, 20.29%, 14.99%, 14.86%, 14.05%, and 13.36%. This result demonstrates that the contributions of all six dictionary atoms to the recovery of the original samples are comparable. For Cessna Citation S/II, the number of spikes in each learned dictionary atom is very small, as depicted in Fig. 14(d), and the ARREs of dictionary atoms are 60.56%, 18.97%, 6.03%, 5.69%, 4.76% and 3.99%, respectively, which indicates that the observed samples are mainly described by two dictionary atoms. For An-26, the number of spikes in each learned dictionary atom is also small, as depicted in Fig. 14(f), and four dictionary atoms are mainly used for An-26 to describe the observations (the ARREs of the dictionary atoms are 30.82%, 20.21%, 18.22%, 13.94%, 7.72%, and 6.31%). Therefore, it is reasonable to conclude that our model is capable of extracting structural characteristics of data.

The dictionary atoms’ property of representing the basic structures of data makes our model learn only a few dictionary atoms. In Fig. 15, we compare the optimal number of dictionary atoms in

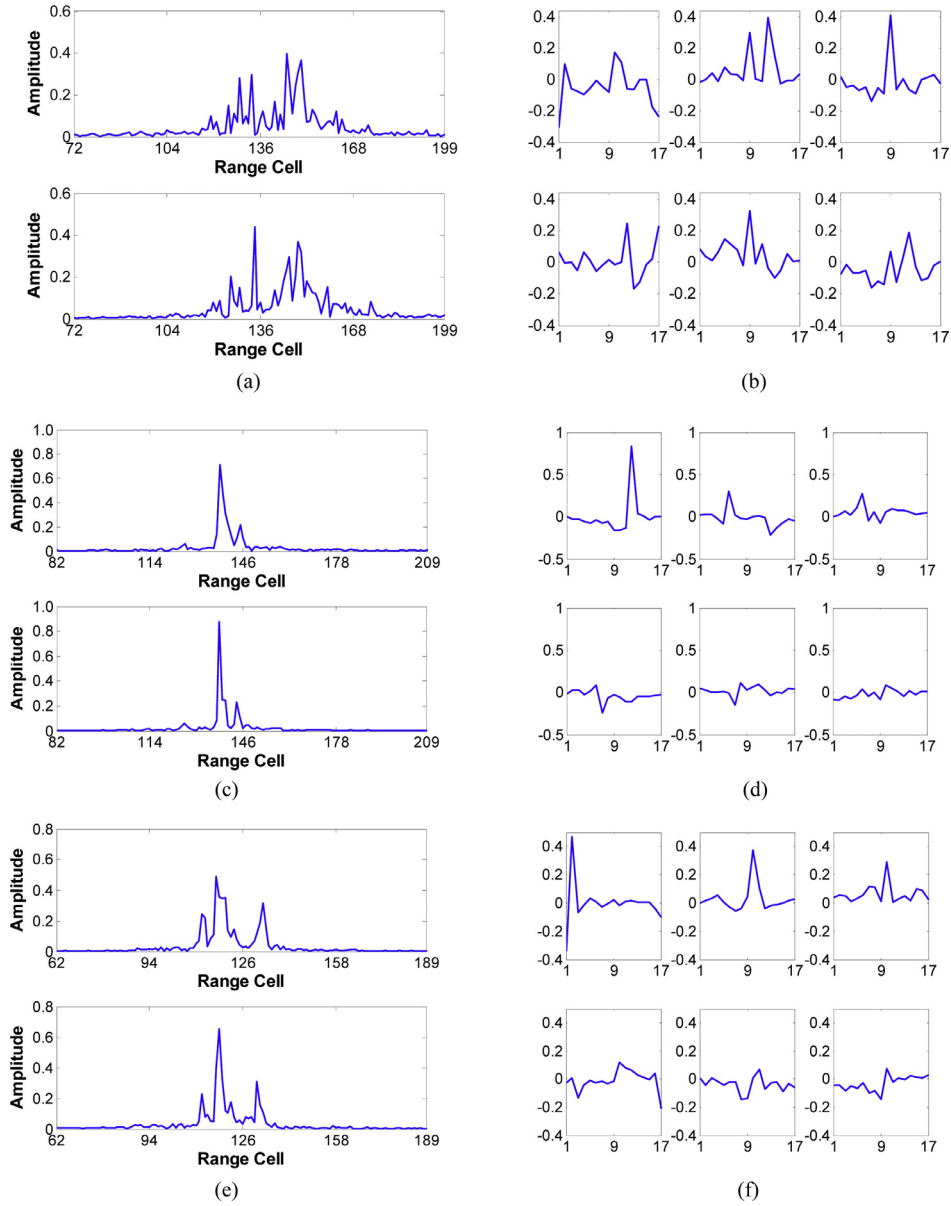


Fig. 14. HRRP samples and the learned dictionary atoms of three frames, with each frame corresponding to a different target. (a), (c) and (e) show two HRRP samples from Yark-42, Cessna Citation S/II, and An-26, respectively, and only the HRRPs with 128 range cells close to the support are shown here for clarity. (b), (d), and (f) present the learned dictionary atoms of Yark-42, Cessna Citation S/II, and An-26, respectively, where the ARREs of the dictionary atoms for each aspect-frame decrease from left to right and top to bottom.

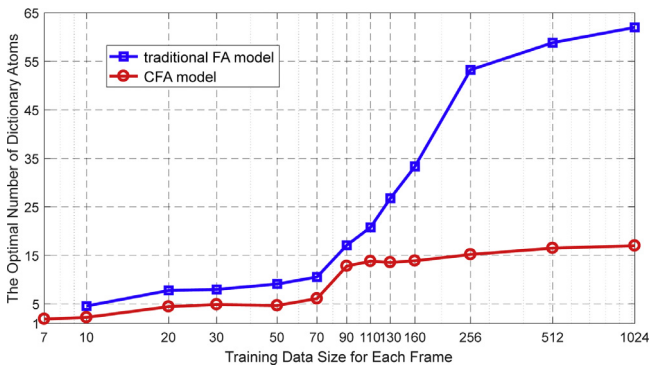


Fig. 15. Comparison of the optimal number of the dictionary atoms in the CFA model and the traditional FA model with different training data sizes for each frame. The results shown here are averaged over 10 runs.

the CFA model and the traditional FA model with different training data sizes for each frame (the method to determine the optimal number of the dictionary atoms for the traditional FA model is similar to that used for the CFA model mentioned in Section 4.2.1). As shown in Fig. 15, the traditional FA model needs to learn more dictionary atoms than the CFA model, especially when the training sample number is larger than 90. It should be noted that the optimal number of the dictionary atoms in the traditional FA model is smaller than that in the CFA model when the training data size for each frame is 7. However, the traditional FA model almost loses its recognition capability in such a condition. Therefore, we omit the blue curve corresponding to the number of training samples per frame smaller than 10 in Fig. 15 for a reasonable comparison.

4.2.4. Analysis on the dimension of dictionary atom

The viewpoint that the lower dimension of the dictionary atom results in the less number of the dictionary atoms is well verified

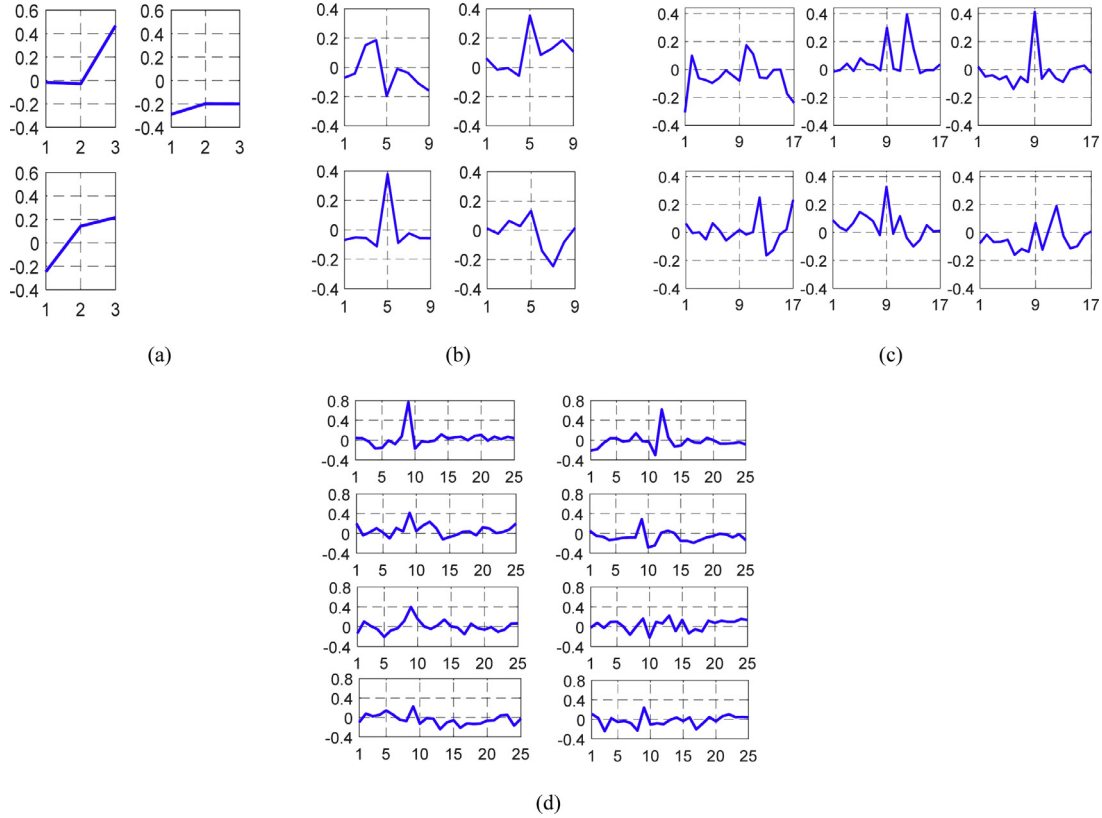


Fig. 16. The inferred dictionaries from an aspect-frame of Yark-42 via the CFA model with different dictionary atom dimensions. There are 70 HRRP samples in this frame and the dimensions of the dictionary atom in (a), (b), (c), and (d) are 3, 9, 17, and 25, respectively.

by the experiments based on synthetic data. We next validate this relationship by using the measured HRRP data.

In this experiment, the CFA models with dictionary atom sizes of 3×1 , 9×1 , 17×1 , and 25×1 , i.e., $J = 3$, $J = 9$, $J = 17$, and $J = 25$, are trained by using 70 HRRP samples from an aspect-frame of Yark-42. The learned dictionaries from one experiment are depicted in Fig. 16. Note that the structural information is considerably simple when $J = 3$, and only three dictionary atoms are required to represent data. In the case of $J = 9$, the inferred dictionary atoms contain a small number of spikes, and their local shared characteristics are weakened. As a result, more dictionary atoms (four dictionary atoms) need to be learned. With the increase of dictionary atom dimension, the number of spikes that appear in a dictionary atom and the number of atoms that need to be learned are increased, as shown in Fig. 16(c) and (d).

In Fig. 17, we present the optimal number of dictionary atoms in the CFA models with different atom dimensions versus the size of training data. As we see from Fig. 17, the CFA model needs to learn an increased number of dictionary atoms when the dimension of the dictionary atom increases, especially when the training data size per frame is larger than 90.

From the experiments on synthetic and measured data, we conclude that the lower dimension of the dictionary atom leads to a smaller number of dictionary atoms to be learned, hence a fewer parameters in the CFA model. In the case of a little observed data, the CFA model with the lower dictionary atom dimension, i.e., a fewer model parameters, can describe the distribution of observations better (a more precise description of the covariance matrix of data). Nevertheless, when the observed data increases, the ability of the CFA model with the lower dictionary atom dimension to depict data becomes poor due to the under-fitting for observations,

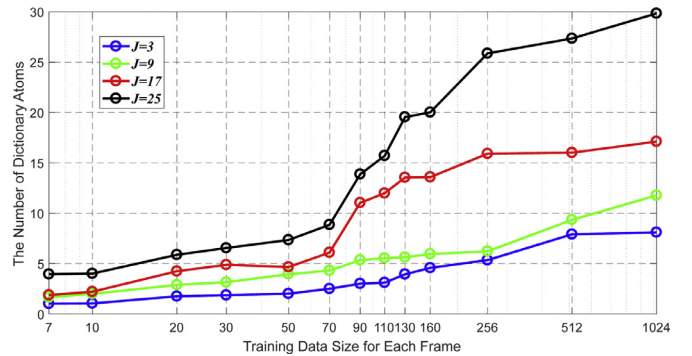


Fig. 17. Optimal number of dictionary atoms in the CFA model with different dictionary atom dimensions versus the size of training data. The results are averaged over 10 model runs.

and subsequently, the recognition performance of the CFA model is limited.

The recognition accuracies of the CFA models with different dictionary atom dimensions, versus the training data size are shown in Fig. 18. From Fig. 18, we derive the following points.

- When the number of training data per frame is less than 10, the recognition performance of the CFA models with $J = 3$ and $J = 9$ is slightly better than that of the CFA with $J = 17$. However, their recognition rates are lower when the training data size for each frame is larger than 10 and nearly unchanged in the case of training sample number larger than 160;
- Since the CFA model with $J = 25$ has the largest number of model parameters, its recognition performance is the worst

Table 2

The total numbers of free parameters for the traditional FA model, subspace approximation model, PPCA model, MTL-FA model, and CFA model.

Models	The total number of free parameters
Traditional FA model	$\sum_{c=1}^C \sum_{m=1}^{M_c} (PV_{opt} + P + 1)$ V_{opt} denotes the optimal number of dictionary atoms for each frame in the FA model
Subspace approximation model	$\sum_{c=1}^C \sum_{m=1}^{M_c} (\frac{P^2+P}{2} + P)$
PPCA model	$\sum_{c=1}^C \sum_{m=1}^{M_c} (\frac{P^2+P}{2} + P)$
MTL-FA model	$PK_s + (P + 1) \sum_{c=1}^C M_c$ K_s denotes the number of the shared factors
CFA model	$\sum_{c=1}^C \sum_{m=1}^{M_c} (JK_{opt} + P + 1)$ K_{opt} denotes the optimal number of dictionary atoms for each frame in the CFA model

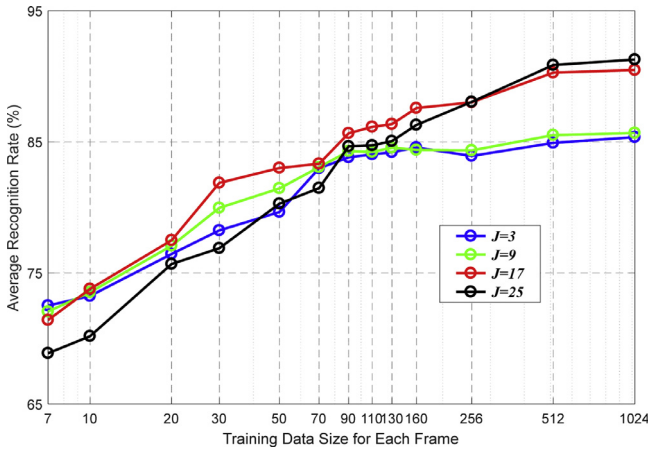


Fig. 18. Recognition accuracies of the CFA model with different dictionary atom dimensions versus the training data size. All results are averaged over 10 runs.

under the small sample size condition. However, its recognition rate is the highest when the training samples are sufficient (the training data size per frame is between 256 and 1024).

Consequently, for our measured HRRP data, the dimension of the dictionary atom being 17 is a better choice.

4.2.5. Comparison of related models

In this experiment, we will evaluate the performance of our CFA model in terms of model complexity and recognition rate by comparing it with the traditional FA model, subspace approximation model [21], PPCA model [21], and MTL-FA model [17].

Table 2 shows the expressions of the total number of free parameters for each model. To describe the model complexity intuitively, the total numbers of free parameters for the five models, versus the size of the training data are shown in Fig. 19. Here, the number of the dictionary atoms in the CFA model and traditional FA model are both optimal. The number of the shared factors in the MTL-FA model, i.e., K_s , is inferred by using the Beta Process (BP) structure [17]. In addition, for the MTL-FA model, the HRRP samples from all frames of all targets are used simultaneously, which requires a large computer storage space. Due to the limitation of our computer memory, the model complexity and recognition rate are no longer considered for the MTL-FA model when the training data size is larger than 130 per frame.

As shown in Fig. 19, the total numbers of free parameters in the subspace approximation model and PPCA model are the largest and does not change with the training data size of each frame be-

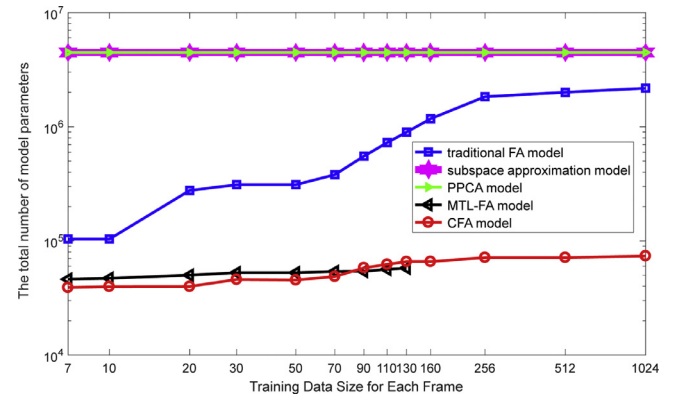


Fig. 19. Variation of the total numbers of free parameters with training data size, via the traditional FA model, subspace approximation model, PPCA model, MTL-FA model, and our CFA model. Due to the limitation of our computer's memory, the total number of free parameters is no longer considered for the MTL-FA model when the training data size is larger than 130 per frame. The results are averaged over 10 runs.

cause regardless of the number of HRRP samples in each frame, the parameter estimations in these two models both depend on the estimation of the $P \times P$ dimensional sample covariance matrix for each frame. Usually, an HRRP sample includes many range cells ($P = 256$ in this paper), therefore, the computation burdens of parameter estimations for these two models are very large. The traditional FA model does not directly estimate the sample covariance matrix, but it estimates the dictionary and covariance matrix of the noise variable. Thus, the number of free parameters in the traditional FA model is smaller than that in the subspace approximation model and PPCA model, as shown in Fig. 19. Moreover, both the MTL-FA model and our CFA model can further reduce the number of model parameters greatly compared with the traditional FA model, and the reasons that they can reduce the parameters have been demonstrated in Section 1. The fewer the model parameters are, the lower the model complexity is, and the fewer training samples are required. Therefore, the MTL-FA model and our CFA model can achieve more accurate parameter estimation with limited training data, followed by the traditional FA model, subspace approximation model and PPCA model.

The recognition accuracies of the five statistical models versus the size of the training data are shown in Fig. 20. As we can see from Fig. 20, the recognition performance of the CFA model is superior to those of the other models when the training sample number of each frame is less than 90. When the training data size per frame is between 90 and 130, the MTL-FA model is slightly better than our CFA model, and both of them outperform the three

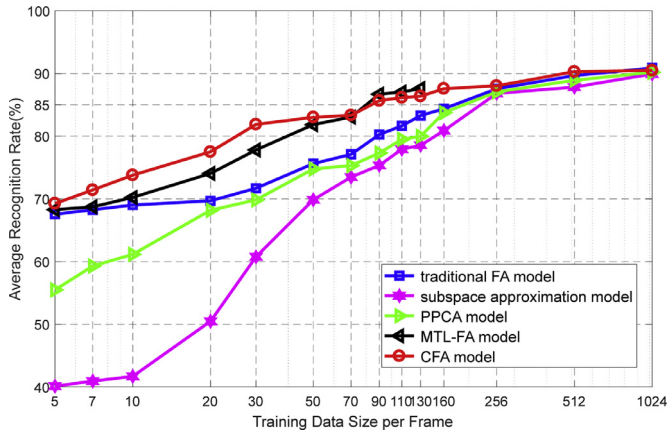


Fig. 20. Variation of the recognition performance with training data size for the traditional FA model, subspace approximation model, PPCA model, MTL-FA model, and CFA model. The recognition rates are averaged over 10 runs.

other models. When the training samples are sufficient, i.e., the number of training samples per frame is larger than or equal to 256, the recognition rates of the traditional FA model, subspace approximation model, PPCA model and CFA model are close and generally remain stable at last. In terms of the recognition rate higher than 85% as a benchmark to determine the number of samples, the MTL-FA model and CFA model only need 90 samples, whereas the three other models need 256 samples. If we assume that the average recognition rate threshold is 70%, our CFA model can work with seven training samples for each frame, however, the MTL-FA model, traditional FA model, PPCA model and subspace approximation model need about 10, 30, 30, and 50 training samples per frame, respectively. Consequently, our CFA model exhibits superior recognition performance with small training data size.

From Figs. 19 and 20, we note that the CFA model has fewer parameters to be estimated and much higher recognition rates than the other methods in the case of training data size per frame smaller than 50. When the training data size per frame is between 90 and 130, the MTL-FA model yields higher recognition accuracies due to its fewer free parameters, and the difference in the recognition performance of the five models decreases, especially for the MTL-FA and CFA models. These results indicate that a smaller number of model parameters can lead to a higher recognition rate when there are only a few training samples. Moreover, as the number of training samples increases, the influence of model parameter number on the recognition performance weakens gradually. We also see in Fig. 20 that the recognition accuracy of the model increases with the increase of training samples because a larger training data size could offer a more accurate parameter estimation and could also provide more information reflecting the targets' characteristics. Therefore, we predict that the recognition rate of the MTL-FA model will continue to increase along with the training data size increasing when the number of training samples per frame is larger than 130. In addition, the recognition accuracy of the MTL-FA model would be lower than that of the CFA model when the number of training samples increases to a certain extent. The reason is that more training samples make it possible for the estimated parameters to accurately represent the observations, and the recognition performance is mainly affected by the statistical models. Compared to the MTL-FA model where the dictionary is shared for all samples from all classes, a CFA model is constructed for each frame of each class. As a result, the dictionary of a frame has the best description for the observed data belonging to this frame, and the dictionaries from different frames are more distinguishable, which helps to achieve higher recognition rate. At

last, the recognition rate of the MTL-FA model should be basically stable, which also occurs in the other models. This is because the training samples are sufficient enough to contain almost all the required information reflecting the true frame-conditional likelihood, and further increments in training data can only provide little information.

To obtain insights into the performance of the models in different targets, we present the confusion matrices of the five models in Fig. 21, where the results of each model are averaged over 10 runs. Some targets of the subspace approximation and PPCA models have higher recognition rates at the expense of the much lower recognition rates of the other targets, and therefore, these two models have the lower average recognition accuracies. For the traditional FA model, the difference in recognition performance among different targets is smaller than that of the subspace approximation and PPCA models, but the recognition rate of each target remains lower; thus, the model's average recognition accuracy is moderate. The increase in the recognition performance of the MTL-FA model primarily originates from the improvement in Cessna Citation S/II, but the accuracies of the two other targets increase slightly. The recognition rates of all targets for our CFA model are similar and relatively high, which leads to the better average recognition rate. Consequently, our CFA model can more accurately describe the diversities among different targets under the small training data size condition.

Aside from the recognition rate, the marginal likelihood of the observed data given the model distribution is another measurement of the performance of the learned model, and it reflects the fitting degree of statistical model and observations. The more accurate the parameter estimation is, the higher the fitting degree of statistical model and observations is, and the greater the marginal likelihood is. In Fig. 22, we present the average marginal log-likelihood of HRRP training samples $\{\mathbf{x}_n^{(c,m)}\}_{n=1, m=1, c=1}^{N, M_c, C}$, i.e., $(\sum_{c=1}^C M_c \cdot N)^{-1} \sum_{c=1}^C \sum_{m=1}^{M_c} \sum_{n=1}^N \ln p_{\Phi^{(c,m)}}(\mathbf{x}_n^{(c,m)})$, in which $\Phi^{(c,m)}$ denotes the model parameter of the m th aspect-frame of the target c , under different training data sizes (different N), via the subspace approximation, PPCA, traditional FA, MTL-FA and CFA models.

From Fig. 22, we note that both our CFA model and the MTL-FA model have higher average marginal log-likelihood than the three other models when the training sample size is small (the number of the training data size for each frame is between 7 and 160), which indicates that our CFA model can obtain more precise parameter estimation. In addition, similar to recognition rate, the marginal likelihood is also affected by the estimation precision of model parameters (mainly in the case of limited training samples) and the statistical model (mainly when abundant training samples are available). Therefore, the average marginal log-likelihood should have the same varying tendency as the recognition rate for the MTL-FA model with training samples per frame larger than 130.

Fig. 19 shows that the total numbers of free parameters for MTL-FA and CFA models are similar. Here the approaches used to reduce parameters are different. The MTL-FA model shares the dictionary atoms, and our CFA model introduces the convolution operation. However, compared to the CFA model which is constructed for each frame of each target, the MTL-FA model uses all training samples from all targets to learn the shared dictionary; therefore, it entails a large computation burden, massive storage space, and excessive training time.

Table 3 presents the average training time of the MTL-FA model and CFA model under different training data sizes, where we take the variation rate of the lower bound lower than 10^{-5} as the end of the iteration. Compared to MTL-FA model, our CFA model consumes less training time. In particular, the training time cost of the CFA model decreases by nearly 99.39% when the training data size

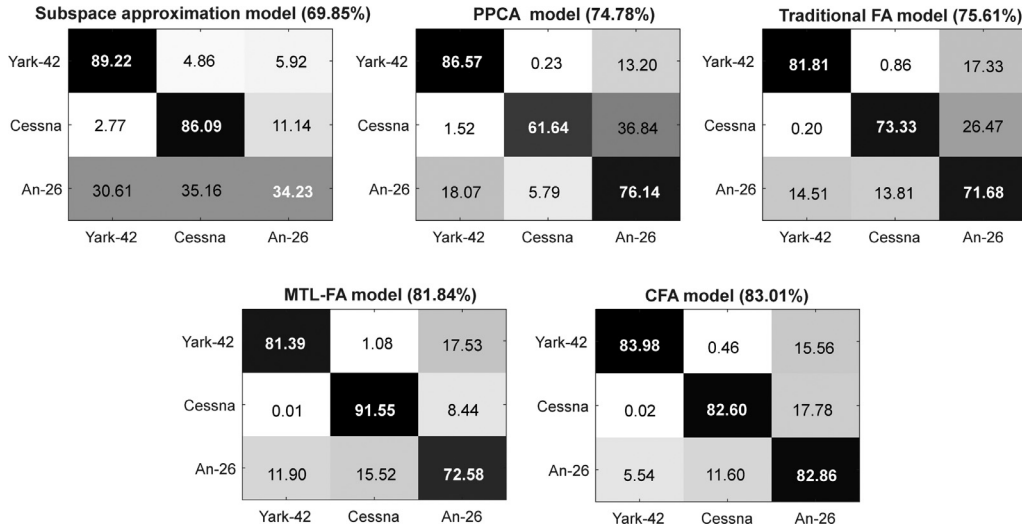


Fig. 21. Confusion matrices of the subspace approximation model, PPCA model, traditional FA model, MTL-FA model, and CFA model, with the training data size per frame being 50. In each subfigure, the number in parentheses in the title is the average recognition rate of the corresponding model. All results shown here are averaged over 10 runs.

Table 3

Average training time of the MTL-FA model and CFA model (min).

Training data size per frame	7	10	20	30	50	70	90	110	130
MTL-FA model	26.8	44.2	123.3	204.6	379.7	533.5	739.2	961.8	1210.7
CFA model	0.7	0.9	1.0	1.4	2.3	6.7	29.0	62.5	101.6

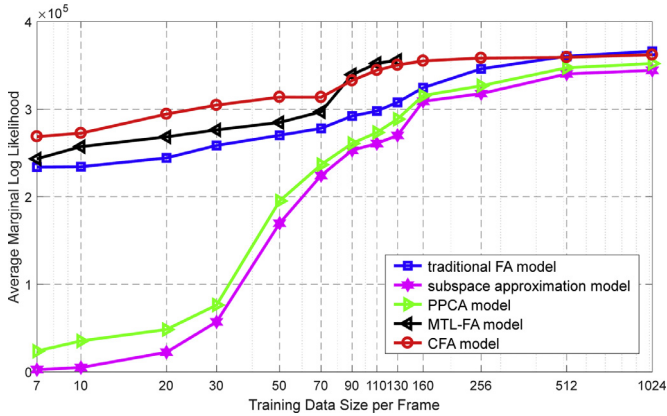


Fig. 22. Comparison of the CFA model to the subspace approximation model, PPCA model, traditional FA model and MTL-FA model, in terms of the average marginal log-likelihood, under different training data sizes. The results shown here are averaged over 10 runs.

per frame is 50. Therefore, our CFA model has higher computing efficiency than MTL-FA model.

5. Conclusions

Statistical machine learning methods have been extensively used in many areas, such as RATR [17,21,24,31–34]. In this paper, a novel statistical model, i.e., CFA model, which is robust to the condition of limited training data, is presented and applied to radar HRRP target recognition. Each dictionary atom in our CFA model is utilized as a convolution kernel with a lower dimension and is capable of extracting the basic structure hidden in data, thus showing the potential to represent the observations with fewer dictionary atoms. Moreover, due to the properties of lower atom dimension and smaller atom number, our proposed method has a much

lower degree of model complexity and can be learned better with limited training data compared with the traditional FA model. Experimental results on synthetic and measured data show that the CFA model can mine the structural information of data and have inspiring recognition performance with a small number of training samples.

In the future, we will mainly focus on investigating multi-layered models for RATR. By using a deep model, the deep structural features reflecting the physical characteristics of data are expected to be extracted, and recognition capability could be ultimately improved.

Acknowledgements

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